The Mathematics of Tatiana

A Cellular–Sheaf Cognitive Architecture over a
Stratified Simplicial Complex

A complete, implementation-free consolidation,
with epistemic status attached to every claim

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How to read this book

What this is

This book consolidates the entire mathematical content of TATIANA into a single document. Its sources are the files in `DOCS/`, the poster sections in `POSTER/`, and the specification documents added on the branch `fix-11-13-and-curvature-decisions`. It is *implementation-free*: no C++, no Python. Where the implementation determines a mathematical fact — and it sometimes does — the fact is stated abstractly here and the discrepancy is recorded in Appendix B.

Why every claim carries a tag

The source documents *contradict one another*. The internal audit (`DOCS/AUDIT_SCRUTINY_AND_BOOK.md`) records fourteen findings against the earlier mathematical description, several of which are outright refutations: a proposition false in the deployed case, a theorem stated more generally than it is proved, a repair for $H^1(\mathcal{C};\mathcal{F})$ that is half circular. A smooth merge of these sources would present refuted material in the same confident register as proved material, and a reader studying it would learn falsehoods.

So every substantive claim in this book carries one of six tags:

[PROVED]	A complete proof is given here, from stated hypotheses.
[PROVED UNDER STATED CONDITIONS]	Proved, but only under hypotheses the deployed system may not
[IMPLEMENTED, UNPROVED]	The system computes this; no proof of correctness exists.
[PROPOSED]	A design intention. Not built, not proved.
[CONTESTED BY AUDIT]	The audit disputes this. The dispute is stated in place.
[REFUTED]	Tested and failed, or shown false. Retained *with* the refutation.

Refuted material is kept rather than deleted. Knowing why a construction failed is most of what you need in order not to rebuild it.

The three layers

Each central object is presented three times over, so that a single reading can be shallow or deep:

1. a boxed **Intuition** in plain language, with no symbols;
2. the formal **Definition**, **Construction** or **Proposition**, with proof;

3. a **Worked example** on an explicit small complex, carried through to numbers.

The worked examples are cumulative. A single running example — a seven-organ complex called K_7, introduced in Chapter 1 — is reused throughout, so that the same object is measured by every instrument the book builds. Where an exercise appears, it is because the step is short and doing it yourself is faster than reading it.

Provenance

Grey boxes marked **Source** record which file a passage descends from, and what was changed in merging it. They exist so that any statement here can be traced back and checked against the original, and so that when the sources disagree you can see *that* they disagree rather than only seeing the winner.

Reading order

Parts I and II are the conceptual spine and everything else depends on them; read them in order. Parts III through VII may be read in any order once the spine is in place. Part VIII is the honest boundary and should be read before quoting any result from this book to anyone else. Parts IX and X are forward-looking: open problems, and the quantum implementation programme.

Contents

Part I

The State Space

Chapter 1

The Base Complex

1.1 What the object is

Intuition. Picture a research assistant that is not one mind but a small society of specialists: one that searches, one that remembers, one that reasons, one that checks. TATIANA is the mathematics of *that society* — who is bound to whom, whether their beliefs agree, by how much they disagree and exactly where, and how the society rewires itself as it works. The reasoning itself is done by an external language model. The mathematics governs the society, not the neurons.

Definition 1.1 (Cognitive state space)**.** A *cognitive state space* is a triple $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$ where

(i) $\mathcal{C}$ is a stratified finite abstract simplicial complex (Section 1.3);

(ii) $\mathcal{F}$ is a cellular sheaf of finite-dimensional real inner-product spaces on $\mathcal{C}$ (Chapter 2);

(iii) $s \in C^0(\mathcal{C}; \mathcal{F})$ is a distinguished 0-cochain, the *current state*, whose coherence is measured by the sheaf Laplacian (Chapter 3).

The space evolves in discrete time under structural operators drawn from an operator algebra $\mathfrak{D}$ which is itself permitted to grow (Part V).

Source. `DOCS/TATIANA_Mathematical_Description.tex`, Def. 1.1, merged with `DOCS/TATIANA_LOGBOOK.md` §0.1. The logbook writes the triple as (C, F, s) with C "a stratified simplicial complex (coarse level = cognitive organs as vertices; fine level = concepts inside each organ's stalk)". Note that this phrasing conflates two different things — the fine complex $\mathcal{C}_v$ and the stalk $\mathcal{F}(v)$ — which are kept strictly separate here and in Chapter 2. See Appendix B, item D1.

1.2 Simplicial complexes and n-ary binding

Intuition. A graph records *pairwise* facts: A relates to B. But a theorem depending on three hypotheses is *one* three-way fact, not three separate pairs — drop any one hypothesis and the fact is gone, which is not how three independent edges behave. A simplicial complex is the smallest honest language for facts of arbitrary arity: a filled triangle is a genuine three-way binding.

Definition 1.2 (Abstract simplicial complex)**.** Let V be a finite set. An *abstract simplicial complex* $\mathcal{C}$ on V is a family of nonempty subsets of V, called *simplices*, which is *downward closed*: if $\sigma \in \mathcal{C}$ and $\emptyset \neq \tau \subseteq \sigma$, then $\tau \in \mathcal{C}$. A simplex of cardinality $k+1$ is a *k-simplex*; write $\dim \sigma = k$. If $\tau \subseteq \sigma$ we write $\tau \trianglelefteq \sigma$ and call τ a *face* of σ. The set of k-simplices is denoted $\mathcal{C}_k$.

Thus 0-simplices are vertices, 1-simplices edges, 2-simplices filled triangles, and so on. Downward closure is not a technical convenience; it is the modelling assumption that every subcollection of a jointly bound fact is itself recorded.

Remark 1.3 (Simplices are n-ary relations)*.* A k-simplex $\{v_0, \ldots, v_k\}$ models a single $(k+1)$-ary binding among its vertices, not $\binom{k+1}{2}$ pairwise ones. This distinction is what later earns the use of b_1 as a diagnostic (Part IV): a 1-cycle counts as a genuine structural hole only because a *filled* 2-simplex means "these three are jointly bound" rather than merely "each pair is bound".

Remark 1.4 (Two relations on the same vertices)*.* A logical prerequisite chain is *directed*: A is needed for B. A simplicial complex is undirected. TATIANA therefore carries two relations on the same vertex set — a directed poset for logical dependency, and the undirected simplices for n-ary co-binding. They are never conflated, and no result in this book uses one where the other is meant.

1.3 The two-level stratification

Intuition. The society has two scales. Zoom out and the vertices are whole *organs*. Zoom into one organ and it is itself a little complex whose vertices are *concepts*. The same geometry runs at both scales, and the two are glued by a coarse-graining map developed in Part III.

Construction 1.5 (The stratified complex)**.** $\mathcal{C}$ carries two strata.

- The *coarse complex* K has as vertices the cognitive organs

  $$V(K) = \{\textsc{Planner}, \textsc{Search}, \textsc{Canonicalizer}, \textsc{Curator}, \textsc{WorkingMemory}, \textsc{Reasoning}, \textsc{Verify}\}.$$

  A simplex $\sigma \in K$ is a *bound coalition* of organs acting as a unit, carrying a time-dependent coupling weight $w(\sigma, t) \in [0, 1]$.
- For each organ $v \in V(K)$, a *fine complex* $\mathcal{C}_v$ has concepts as vertices and n-ary conceptual bindings as simplices. This is the organ's internal knowledge.

The stratification answers an ambiguity present in the earliest drafts — *are the vertices concepts or subsystems?* — by answering *both*, at two different levels.

Claims and non-claims. [CONTESTED BY AUDIT] The word "stratified" is used here in a weak, combinatorial sense: two nested scales with a map between them. It does *not* refer to a Whitney stratification, a stratified space in the sense of intersection homology, or a constructible sheaf on a stratified base, and no theorem from those theories is invoked anywhere in this book. Earlier drafts borrowed vocabulary from that literature; the logbook (§0.5) records that most such borrowing was cut as decorative. This is a naming collision, and the reader should not import expectations from it.

1.4 The running example

Every instrument built in this book will be applied to one small complex, so that the same object is measured repeatedly and the measurements can be compared.

Example 1.6 (The complex K_4)**.** Let K_4 have vertex set

$$V = \{\mathsf{S},\ \mathsf{M},\ \mathsf{R},\ \mathsf{V}\}$$

(read: Search, Memory, Reasoning, Verify), with 1-simplices

$$E = \big\{\{\mathsf{S},\mathsf{M}\},\ \{\mathsf{M},\mathsf{R}\},\ \{\mathsf{S},\mathsf{R}\},\ \{\mathsf{R},\mathsf{V}\}\big\}$$

and a single 2-simplex $\{\mathsf{S},\mathsf{M},\mathsf{R}\}$, filled. Downward closure holds: all three faces of the triangle are in E.

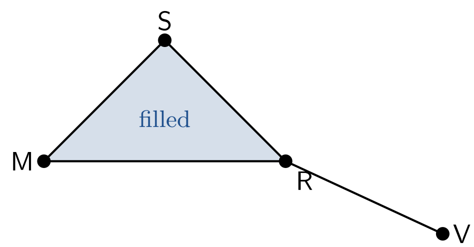

Figure 1.1: The running example K_4. The triangle $\{\mathsf{S},\mathsf{M},\mathsf{R}\}$ is filled: it is one ternary binding, not three binary ones. The edge to V is a pendant.

Its degrees are $\deg \mathsf{S} = 2$, $\deg \mathsf{M} = 2$, $\deg \mathsf{R} = 3$, $\deg \mathsf{V} = 1$, and its graph Laplacian, in the vertex order $(\mathsf{S},\mathsf{M},\mathsf{R},\mathsf{V})$, is

$$L_{K_4} = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \tag{1.1}$$

Proposition 1.7 (Spectrum of the running example)**.** **[PROVED]** *The characteristic polynomial of L_{K_4} is $\lambda(\lambda-1)(\lambda-3)(\lambda-4)$, so*

$$\operatorname{spec} L_{K_4} = \{0,\ 1,\ 3,\ 4\}.$$

Proof. Direct expansion of $\det(L_{K_4} - \lambda I)$ gives $\lambda^4 - 8\lambda^3 + 19\lambda^2 - 12\lambda = \lambda(\lambda-1)(\lambda-3)(\lambda-4)$. The eigenvalue 0 is forced: $L_{K_4}\mathbf{1} = 0$ since every row sums to zero. Its multiplicity is 1 because the 1-skeleton is connected (Proposition 3.6). □

The integrality here is a happy accident of this particular graph and should not be expected in general. It is why K_4 was chosen: every quantity in Parts I–IV can be carried through in exact arithmetic.

Chapter 2

The Sheaf

> **Intuition.** On top of the shape we lay *data*. Each organ holds a vector describing what it currently believes. Each binding holds a space in which the beliefs of its members are meant to be compared. A sheaf is exactly this: local data on every piece, plus rules for moving data into the places where comparison happens. A "coherent thought" is then a choice of data that agrees everywhere it is compared.

2.1 Which way do the restriction maps point?

Before the definition, one question must be settled, because the source documents answer it in two incompatible ways and the answer determines everything downstream.

Given a face relation $\sigma \trianglelefteq \tau$ — say a vertex u on an edge e — a sheaf must supply a linear map between $\mathcal{F}(u)$ and $\mathcal{F}(e)$. Which direction?

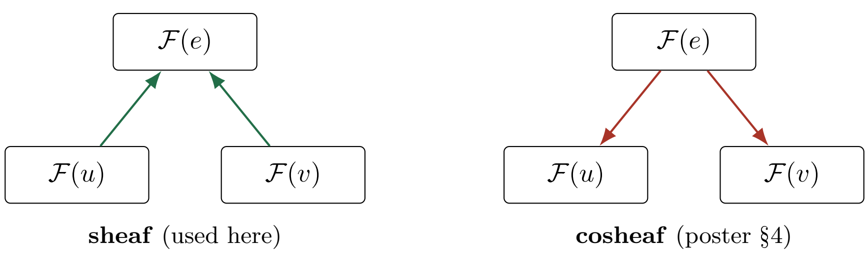

Figure 2.1: The two possible orientations. Only the left one makes $\ker \delta$ the space of agreeing assignments.

Proposition 2.1 (The direction is forced)**.** **[PROVED]** *Suppose we want a linear operator δ from vertex data to edge data whose kernel is exactly the set of assignments $x = (x_v)_{v \in V}$ that "agree on every edge". Then the structure maps must go* from faces to cofaces*, i.e. $\mathcal{F}_{u \trianglelefteq e} : \mathcal{F}(u) \to \mathcal{F}(e)$.*

Proof. An assignment gives one vector $x_v \in \mathcal{F}(v)$ per vertex, and nothing on edges. For the two endpoint values x_u, x_v of an edge $e = \{u, v\}$ to be *compared* at all, they must be brought into a common space. The only space available to e is $\mathcal{F}(e)$. Hence we require maps $\mathcal{F}(u) \to \mathcal{F}(e)$ and $\mathcal{F}(v) \to \mathcal{F}(e)$, and agreement on e means $\mathcal{F}_{u \trianglelefteq e}(x_u) = \mathcal{F}_{v \trianglelefteq e}(x_v)$.

Suppose instead the maps ran downward, $\rho_{e \to u} : \mathcal{F}(e) \to \mathcal{F}(u)$. Then from vertex data alone no map into $\mathcal{F}(e)$ exists, so no operator $C^0(\mathcal{C}; \mathcal{F}) \to C^1(\mathcal{C}; \mathcal{F})$ can be built from the structure maps, and the condition "x agrees on e" is not expressible. Downward maps instead assemble into a boundary operator $\partial : C^1(\mathcal{C};\mathcal{F}) \to C^0(\mathcal{C};\mathcal{F})$, whose kernel consists of *edge* data summing to zero at vertices — a flow condition, not an agreement condition. That is the defining structure of a *cellular cosheaf*, and its invariants are homology $H_\bullet$, not the cohomology $H^\bullet$ used throughout this book. □

Claims and non-claims. [REFUTED] The poster section `POSTER/sections/05_sheaf.tex` defines, for $\sigma \subseteq \tau$, a map $\rho_{\tau\to\sigma} : \mathcal{F}(\tau) \to \mathcal{F}(\sigma)$, and its diagram draws $\mathcal{F}(e)$ above the vertices with arrows pointing down. It then asserts, three boxes later, that $s \in H^0(\mathcal{C};\mathcal{F})(\mathcal{C},\mathcal{F})$ is a global section. By Proposition 2.1 these two statements are incompatible: with downward maps the object defined is a cosheaf and $H^0(\mathcal{C};\mathcal{F})$ as "the space of compatible assignments" does not arise. The main mathematical description (`DOCS/TATIANA_Mathematical_Description.tex`, Def. 3.1) has the direction correct. The poster is wrong and should be corrected before it is shown to anyone. Logged as Appendix B, item D2.

Exercise 2.2. Take the downward convention seriously for one minute and try to build $\delta : C^0(\mathcal{C};\mathcal{F}) \to C^1(\mathcal{C};\mathcal{F})$ from it. Where exactly does the construction fail — at the definition of δ, or later at the claim $\ker\delta = \{\text{agreeing assignments}\}$? Answering this is the fastest way to internalise why the direction is not a convention.

2.2 The definition

Definition 2.3 (Cellular sheaf)**.** A *cellular sheaf* $\mathcal{F}$ of real inner-product spaces on a simplicial complex $\mathcal{C}$ assigns

(i) to each simplex $\sigma \in \mathcal{C}$ a finite-dimensional real inner-product space $\mathcal{F}(\sigma)$, the *stalk* over σ;

(ii) to each face relation $\sigma \trianglelefteq \tau$ a linear *restriction map*

$$\mathcal{F}_{\sigma\trianglelefteq\tau} : \mathcal{F}(\sigma) \longrightarrow \mathcal{F}(\tau),$$

subject to functoriality: $\mathcal{F}_{\sigma\trianglelefteq\sigma} = \mathrm{Id}$, and for every chain $\sigma \trianglelefteq \tau \trianglelefteq \upsilon$,

$$\mathcal{F}_{\tau\trianglelefteq\upsilon} \circ \mathcal{F}_{\sigma\trianglelefteq\tau} \;=\; \mathcal{F}_{\sigma\trianglelefteq\upsilon}. \tag{2.1}$$

Equivalently, $\mathcal{F}$ is a functor from the face poset of $\mathcal{C}$, viewed as a category, to $\mathbf{Vect}_{\mathbb{R}}$.

Remark 2.4 (Functoriality is a real constraint)*.* Condition (2.1) is vacuous on a complex of dimension ≤ 1, since there are no chains of length two. It becomes a genuine constraint the moment a 2-simplex is filled — which K_4 does. Any procedure that manufactures restriction maps edge-by-edge (for instance by fitting each $\mathcal{F}_{v\trianglelefteq e}$ independently to data) will in general *violate* (2.1) on filled triangles, and the resulting object is then not a sheaf. This failure mode is not hypothetical; it is the standard obstruction encountered when L^2-projections are used as restriction maps, and the canonical repair is to promote the offending interface to a cell of its own so that the composite is not required to commute. Part VI returns to this.

Definition 2.5 (Cochains)**.** For $k \geq 0$ set

$$C^k(\mathcal{C};\mathcal{F}) \;=\; \bigoplus_{\sigma\in\mathcal{C}_k} \mathcal{F}(\sigma),$$

the space of *k-cochains*, with the direct-sum inner product $\langle x, y\rangle = \sum_\sigma \langle x_\sigma, y_\sigma\rangle$. An element of $C^0(\mathcal{C}; \mathcal{F})$ is an assignment of one vector to each organ; the current state s of Definition 1.1 lives here.

2.3 What the stalks contain

Construction 2.6 (Deployed stalk data)**. [IMPLEMENTED, UNPROVED]** In the deployed system every vertex and edge stalk is $\mathbb{R}^d$ with $d = 384$, and the vector $x_v \in \mathcal{F}(v)$ is a sentence embedding produced by a fixed local encoder. Two invariants are enforced at the level of types:

(i) dimensions never truncate or pad — a mismatch raises rather than silently reshaping;

(ii) an idle organ carries *no* stalk value, a formal $\perp$, never the zero vector.

Claims and non-claims. Invariant (ii) is the sheaf-theoretic content of "do not fabricate data", and it is worth stating why it is not pedantry. The zero vector is a genuine point of $\mathbb{R}^d$: it has a distance to every other point, it contributes to $\|X\|_F^2$, and it participates in every average. Substituting 0 for "unknown" therefore does not represent absence, it asserts a specific belief — that the organ believes the origin — and drags every downstream distance toward it. The distinction between $\perp$ and 0 is the difference between a measurement not made and a measurement made and found to be zero.

Remark 2.7 (Where the fine complex lives). The logbook occasionally describes concepts as living *inside an organ's stalk*. This is loose. The stalk $\mathcal{F}(v)$ is a 384-dimensional real vector space; the fine complex $\mathcal{C}_v$ is a simplicial complex whose vertices are concepts. They are different objects, and the relation between them is the coarse-graining map π_v of Part III, which sends the concept data of $\mathcal{C}_v$ to a single vector in $\mathcal{F}(v)$. Conflating them makes π_v look like an identity when it is in fact a lossy precision-weighted fusion.

Chapter 3

Coboundary and Laplacian

3.1 The coboundary

Definition 3.1 (Coboundary in degree zero)**.** Fix an arbitrary orientation of each edge, writing $e = (u, v)$ for the edge $\{u, v\}$ oriented from u to v. The *coboundary* $\delta : C^0(\mathcal{C}; \mathcal{F}) \to C^1(\mathcal{C}; \mathcal{F})$ acts edgewise by

$$(\delta x)_e \;=\; \mathcal{F}_{v \trianglelefteq e}(x_v) \;-\; \mathcal{F}_{u \trianglelefteq e}(x_u) \;\in\; \mathcal{F}(e). \tag{3.1}$$

A 0-cochain x is a *global section* if $\delta x = 0$, that is, if every bound pair agrees after being mapped into its shared comparison space.

Lemma 3.2 (Orientation-independence)**.** **[PROVED]** *Reversing the orientation of an edge e replaces $(\delta x)_e$ by its negative. Consequently* $\ker \delta$*, the quantity $\|\delta x\|^2$, and the operator $\delta^\top \delta$ are all independent of the chosen orientations.*

Proof. Reversal exchanges the roles of u and v in (3.1), negating the entry. Negating a coordinate leaves its vanishing, its squared norm, and hence the sum of squared norms unchanged. Writing $\delta' = D\delta$ with D diagonal with entries ± 1 blockwise, we get $\delta'^\top \delta' = \delta^\top D^\top D \delta = \delta^\top \delta$ since $D^\top D = I$. □

Remark 3.3 (Why an arbitrary choice is harmless)*.* Lemma 3.2 is what licenses the phrase "fix an arbitrary orientation". It is worth checking rather than assuming, because the analogous statement fails for the *signed* per-edge quantity $(\delta x)_e$ itself, which is only defined up to sign. Any diagnostic that reports $(\delta x)_e$ rather than $\|(\delta x)_e\|^2$ is reporting an orientation-dependent quantity and is therefore not well defined.

3.2 The sheaf Laplacian

Definition 3.4 (Sheaf Laplacian and discord)**.** The *degree-zero sheaf Laplacian* is

$$L \;=\; \delta^\top \delta \;:\; C^0(\mathcal{C}; \mathcal{F}) \to C^0(\mathcal{C}; \mathcal{F}), \tag{3.2}$$

and the *discord* of a state x is the Dirichlet energy

$$\omega(x) \;=\; x^\top L x \;=\; \|\delta x\|^2 \;=\; \sum_{e=\{u,v\}} \left\|\mathcal{F}_{v \trianglelefteq e}(x_v) - \mathcal{F}_{u \trianglelefteq e}(x_u)\right\|^2. \tag{3.3}$$

Proposition 3.5 (Basic properties)**.** **[PROVED]** *L is symmetric positive semidefinite, and*

$$\omega(x) = 0 \iff \delta x = 0 \iff x \in \ker L.$$

Moreover $\ker \delta = H^0(\mathcal{C}; \mathcal{F})$*, the zeroth sheaf cohomology.*

Proof. Symmetry: $(\delta^\top \delta)^\top = \delta^\top \delta$. Positive semidefiniteness: $x^\top L x = \|\delta x\|^2 \geq 0$. If $Lx = 0$ then $\|\delta x\|^2 = x^\top L x = 0$, so $\delta x = 0$ since $\|\cdot\|$ is a norm; conversely $\delta x = 0$ gives $Lx = 0$ immediately. Finally the cochain complex begins in degree 0 — there are no (-1)-cochains — so $H^0(\mathcal{C}; \mathcal{F}) = \ker \delta^0 / \operatorname{im} \delta^{-1} = \ker \delta$. □

Intuition. ω is a single number: the total squared disagreement across all bindings. It vanishes exactly when the society holds one coherent thought. The point of the construction is that "a coherent thought" has become a precise object — a vector in the kernel of a matrix — rather than a mood.

Proposition 3.6 (Kernel and components)**.** **[PROVED]** *If every restriction map is the identity on a common stalk $\mathbb{R}^d$, then*

$$L = L_K \otimes I_d,$$

where L_K is the ordinary combinatorial graph Laplacian of the 1*-skeleton. Consequently* $\operatorname{spec} L$ *is* $\operatorname{spec} L_K$ *with each eigenvalue repeated d times, and* $\dim \ker L = d \cdot b_0$*, where b_0 is the number of connected components.*

Proof. With identity restrictions (3.3) reads $\omega(x) = \sum_{\{u,v\} \in E} \|x_u - x_v\|^2$, which is the Dirichlet form of L_K applied coordinatewise in each of the d components. Hence $L = L_K \otimes I_d$. The spectrum of a Kronecker product with the identity is the spectrum of the factor with multiplicity d. The kernel of L_K is spanned by the indicator vectors of the connected components, of which there are b_0; tensoring with $\mathbb{R}^d$ multiplies the dimension by d. □

Claims and non-claims. **[IMPLEMENTED, UNPROVED]** The deployed system uses identity restriction maps, so Proposition 3.6 describes the regime it actually operates in. This is honest and cheap, but it is the *crudest* possible sheaf: on a connected 1-skeleton, $\ker L$ is the diagonal $\{x : x_u = x_v \ \forall u, v\}$, so the only coherent states are exact consensus and reconciliation is plain averaging. Everything interesting that a sheaf can express — that two organs agree *after* a change of basis, or agree only on a subspace — lives in non-identity restriction maps. Promoting them (learned $\mathcal{F}_{v \trianglelefteq e}$, for instance by closed-form Procrustes alignment) is the identified route from a diagnostic sheaf to a computational one. It is a direction, not a result.

3.3 The running example, computed

Worked example. Take K_4 from Example 1.6 with identity restrictions and stalk dimension $d = 2$, and give it the state

$$x_{\mathsf{S}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_{\mathsf{M}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_{\mathsf{R}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad x_{\mathsf{V}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Read it as: SEARCH and MEMORY agree with each other, REASONING and VERIFY agree with each other, and the two camps are orthogonal.

Discord, edge by edge. Using (3.3) with identity maps, $\omega_e = \|x_u - x_v\|^2$:

$$\begin{array}{lll} \{\mathsf{S},\mathsf{M}\}: & \|(0,0)\|^2 & =0, \\ \{\mathsf{M},\mathsf{R}\}: & \|(1,-1)\|^2 & =2, \\ \{\mathsf{S},\mathsf{R}\}: & \|(1,-1)\|^2 & =2, \\ \{\mathsf{R},\mathsf{V}\}: & \|(0,0)\|^2 & =0, \end{array} \qquad \Longrightarrow \qquad \omega(x)=4.$$

Localisation. The two guilty edges tie at $\omega_e = 2$. The $\arg\max$ is therefore *not unique*, and a diagnostic that returns a single "worst edge" here is returning an artefact of tie-breaking order, not a measurement. The honest return is the tied set $\{\{\mathsf{M},\mathsf{R}\},\{\mathsf{S},\mathsf{R}\}\}$ — which, read back, correctly says that R is the organ at odds with the rest.

Kernel projection. The 1-skeleton is connected, so by Proposition 3.6 the kernel is the diagonal and x_0 is the constant assignment at the mean $\bar{x} = \frac{1}{4}\big((1,0)+(1,0)+(0,1)+(0,1)\big) = \big(\frac{1}{2},\frac{1}{2}\big)$. Hence

$$\|x_0\|^2 = 4\cdot\left(\tfrac{1}{4}+\tfrac{1}{4}\right) = 2, \qquad x_\perp = x - x_0 = \left(\pm\tfrac{1}{2},\mp\tfrac{1}{2}\right)_v, \qquad \|x_\perp\|^2 = 4\cdot\tfrac{1}{2} = 2,$$

and Pythagoras checks: $\|X\|_F^2 = 2+2 = 4$. □

These numbers are picked up again in Part II, where they are fed to the coherence score ρ and its factorisation, and in Part IV, where the same complex is used to compute b_0, b_1 and $H^1(\mathcal{C};\mathcal{F})$.

Exercise 3.7. Recompute $\omega(x)$ for the same complex after filling in a restriction map that is *not* the identity: let $\mathcal{F}_{\mathsf{R}\trianglelefteq e}$ be rotation by 90° for the two edges at R inside the triangle, and the identity elsewhere. Does ω go up or down, and does the resulting assignment of maps satisfy functoriality (2.1) on the filled triangle $\{\mathsf{S},\mathsf{M},\mathsf{R}\}$? The second half of that question is the important one.

Part II

Coherence

Chapter 4

The Coherence Score

This chapter derives the two numbers the engine computes on every tick, and then shows that one of them is secretly two numbers with different meanings.

4.1 Why a normaliser is needed at all

> **Intuition.** The discord ω has no natural scale. A bigger complex has more edges and therefore a bigger ω for no interesting reason, and a state scaled by a factor of 10 has a discord scaled by 100. What is wanted is a dimensionless thermometer in $[0,1]$: 0 for perfect agreement, 1 for the maximum disagreement the shape permits. The subtlety is getting it *without* an eigendecomposition, because the score is computed on every tick and an eigensolver is not free.

The natural normaliser is $\lambda_{\max}(L)\,\|x\|^2$, since $\omega(x) = x^\top Lx \le \lambda_{\max}(L)\,\|x\|^2$ by the Rayleigh bound. But computing $\lambda_{\max}$ is exactly what we are trying to avoid. The way out is a classical degree bound.

Theorem 4.1 (Anderson–Morley)**.** **[PROVED]** *Let G be a finite simple graph with combinatorial Laplacian L_G. Then*

$$\lambda_{\max}(L_G) \;\le\; \max_{\{u,v\}\in E}\bigl(\deg u + \deg v\bigr) \;=:\; B. \tag{4.1}$$

Proof. Write $L_G = D - A$ and let $Q = D + A$ be the signless Laplacian. Because $Q = NN^\top$ for N the unsigned incidence matrix, Q is positive semidefinite, and $\lambda_{\max}(L_G) \le \lambda_{\max}(Q)$: passing to the bipartite double cover exchanges L_G and Q while preserving the spectral radius, and L_G is a principal restriction of that cover's signless Laplacian.

It therefore suffices to bound $\lambda_{\max}(Q)$. Let y be a unit eigenvector of Q for $\lambda_{\max}$, let u maximise $|y_u|$, and let v be a neighbour of u maximising $|y_v|$. From $\lambda_{\max} y_u = \deg(u) y_u + \sum_{w\sim u} y_w$ we get

$$(\lambda_{\max} - \deg u)\,|y_u| \;\le\; \deg(u)\,|y_v|\,,$$

and symmetrically $(\lambda_{\max} - \deg v)\,|y_v| \le \deg(v)\,|y_u|$. Multiplying the two inequalities and cancelling $|y_u|\,|y_v| > 0$ gives $(\lambda_{\max} - \deg u)(\lambda_{\max} - \deg v) \le \deg u \deg v$, i.e. $\lambda_{\max}^2 - \lambda_{\max}(\deg u + \deg v) \le 0$, whence $\lambda_{\max} \le \deg u + \deg v \le B$. □

Remark 4.2 (The bound is loose, and the looseness costs something)*.* (4.1) is tight only on special graphs — it is exact on complete bipartite graphs. On the running example K_4 it gives

$B = 5$ while $\lambda_{\max} = 4$, a 25% over-estimate. Over-bounding $\lambda_{\max}$ compresses ρ toward zero, which is one of three contributors to the dead dynamic range documented in Section 4.5. Finding a tighter eigensolver-free bound is Open Problem 17.4.

4.2 The score

Definition 4.3 (Coherence score)**.** Let $X \in \mathbb{R}^{|V|\times d}$ stack the active stalk vectors as rows, with Frobenius norm $\|X\|_F$. In the identity-restriction regime the *coherence score* is

$$\boxed{\rho \;=\; \frac{\omega}{B\,\|X\|_F^2}} \tag{4.2}$$

When $E = \emptyset$ or $\|X\|_F = 0$, the score is *undefined* and returns $\perp$ — never 0 and never 1.

Proposition 4.4 ($\rho \in [0, 1]$)**.** **[PROVED UNDER STATED CONDITIONS]** *With identity restriction maps, $0 \le \rho \le 1$.*

Proof. Lower bound: $\omega \ge 0$ and $B\,\|X\|_F^2 > 0$ whenever the score is defined. Upper bound: by Proposition 3.6, $L = L_K \otimes I_d$, so $\lambda_{\max}(L) = \lambda_{\max}(L_K) \le B$ by Theorem 4.1. Hence $\omega = x^\top Lx \le \lambda_{\max}(L)\,\|x\|^2 \le B\,\|X\|_F^2$, using $\|x\| = \|X\|_F$ under the row-stacking identification. □

> **Claims and non-claims.** The hypothesis in Proposition 4.4 is not cosmetic. With *non-identity* restriction maps the Kronecker factorisation fails, $\lambda_{\max}(L)$ is no longer the spectrum of any $|V| \times |V|$ matrix, and B need not bound it. The correct general hypothesis is an operator-norm condition: if every $\|\mathcal{F}_{v\trianglelefteq e}\| \le 1$ then a weighted version of Theorem 4.1 still applies. Without it the score can exceed 1. The recorded design decision is that this case is *reported*, not silently clamped — a value $\rho > 1$ is evidence that the normalisation hypothesis has been violated, and truncating it destroys the only evidence there is.

Remark 4.5 (Why $\perp$ is not 0). An empty edge set means no comparison was made. Returning 0 would report perfect agreement, which is a measurement; returning $\perp$ reports that no measurement exists. The distinction propagates: the controller of Chapter 15 carries a third mode precisely so that "no measurement" does not default to "all is well".

4.3 The score answers two questions at once

> **Intuition.** The numerator ω is *blind* to consensus, because consensus is exactly $\ker L$ and the kernel contributes nothing to the quadratic form. The denominator $\|X\|_F^2$ counts consensus in full. So ρ is disagreement divided by (disagreement plus consensus), tangled together with a second and quite different quantity saying which *shape* the disagreement takes. Untangling them costs one orthogonal projection.

Proposition 4.6 (Exact factorisation)**.** **[PROVED UNDER STATED CONDITIONS]** *Assume identity restrictions. Let $x = x_0 \oplus x_\perp$ be the orthogonal decomposition with x_0 the projection* *onto* $\ker L$*, and suppose* $x_\perp \neq 0$*. Then* $\omega(x) = x_\perp^\top L x_\perp$*,* $\|X\|_F^2 = \|x_0\|^2 + \|x_\perp\|^2$*, and*

$$\boxed{\rho \;=\; \underbrace{\frac{\|x_\perp\|^2}{\|X\|_F^2}}_{\alpha} \cdot \underbrace{\frac{\omega}{B\,\|x_\perp\|^2}}_{\tilde\rho}} \tag{4.3}$$

where α*, the* dissent fraction*, says how much of the state is not consensus, and* $\tilde\rho$*, the* mode index*, says which mode the disagreement occupies.*

Proof. Since $Lx_0 = 0$ and L is symmetric, the cross terms in $x^\top L x = (x_0 + x_\perp)^\top L (x_0 + x_\perp)$ vanish, because $x_0^\top L x_\perp = (Lx_0)^\top x_\perp = 0$. The norm identity is Pythagoras. The factorisation is multiplication and division of (4.2) by $\|x_\perp\|^2$. □

Remark 4.7 (The kernel is per-component, not the global mean). For a connected 1-skeleton, $\ker L$ is the diagonal and $x_0 = \mathbf{1} \otimes \bar{x}$ with $\bar{x}$ the global mean. For $b_0 > 1$ this is *false*: by Proposition 3.6 the kernel consists of assignments constant on *each component*, so x_0 subtracts the **per-component** mean. Using the global mean when $b_0 > 1$ scores between-component separation as dissent, when in fact it lies *inside* the kernel and is not disagreement at all. This is a correctness condition, not an optimisation.

Proposition 4.8 (The achievable window)**. [PROVED UNDER STATED CONDITIONS]** $\tilde\rho = R_L(x_\perp)/B$ *is a Rayleigh quotient of* L *restricted to* $(\ker L)^\perp$*, so by Courant–Fischer*

$$\tilde\rho \;\in\; \Big[\; \lambda^+_{\min}(L_K)/B, \quad \lambda_{\max}(L_K)/B \;\Big], \tag{4.4}$$

where $\lambda^+_{\min}$ *is the smallest* nonzero *eigenvalue — the Fiedler value when* K *is connected. Both endpoints are graph invariants, computable from the* 1*-skeleton before any data is seen, and needing recomputation only when the topology changes.*

Proof. $x_\perp$ ranges over $(\ker L)^\perp$, an invariant subspace on which L is positive definite with spectrum the nonzero eigenvalues. The Rayleigh quotient of a symmetric operator on an invariant subspace attains exactly the closed interval between the extreme eigenvalues there. Dividing by the constant B gives (4.4). □

> **Claims and non-claims.** Four points, in order of importance.
>
> *(i)* (4.3) resolves the non-monotonicity of Remark 15.2: adding a well-aligned organ inflates $\|X\|_F^2$ and barely moves $\|x_\perp\|^2$, so α dilutes while $\tilde\rho$ is untouched. The controller should gate on $\tilde\rho$ and report α alongside.
>
> *(ii)* (4.4) is **normalisation, not derivation**. Setting a threshold ε as a quantile q of the window still requires choosing q. What changes is that ε is expressed in units the graph supplies and adapts automatically to rewiring. It is a magic number made topology-relative, not a magic number eliminated, and it must not be sold as more than that.
>
> *(iii)* The split is exact *only* for identity restrictions. With non-identity maps, $\ker L$ is not the per-component constants and no $|V| \times |V|$ matrix carries the spectrum of L. The correct behaviour is then to report α, $\tilde\rho$ and the window as $\perp$ rather than approximate them. ρ itself is unaffected.

(iv) When $x_\perp = 0$ exactly, $\alpha = 0$ and $\tilde{\rho} = \perp$: perfect consensus has no mode. Returning 0 for $\tilde{\rho}$ would assert "the most global mode possible", a measurement that was not made.

4.4 The running example, continued

Worked example. Continue Example 1.6 with the state of Chapter 3, where we found $\omega = 4$, $\|X\|_F^2 = 4$ and $\|x_\perp\|^2 = 2$.

The degrees are $2, 2, 3, 1$, so

$$B = \max\{2{+}2,\ 2{+}3,\ 2{+}3,\ 3{+}1\} = 5.$$

Hence

$$\rho = \frac{4}{5 \cdot 4} = \frac{1}{5}, \qquad \alpha = \frac{2}{4} = \frac{1}{2}, \qquad \tilde{\rho} = \frac{4}{5 \cdot 2} = \frac{2}{5},$$

and the factorisation checks: $\alpha \cdot \tilde{\rho} = \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{5} = \rho$.

By Proposition 1.7 the nonzero spectrum is $\{1, 3, 4\}$, so

$$\text{window} = \left[\tfrac{1}{5},\ \tfrac{4}{5}\right] = [0.2,\ 0.8],$$

and $\tilde{\rho} = 0.4$ sits at window position $(0.4 - 0.2)/(0.8 - 0.2) = \frac{1}{3}$.

Reading the pair. $\alpha = \frac{1}{2}$ is large: half the state's energy is not consensus. $\tilde{\rho}$ at $\frac{1}{3}$ of its window is low. By the typed reading of Remark 15.3, large α with small $\tilde{\rho}$ is a *two-camp split* — which is exactly how the state was built. The single number $\rho = 0.2$ reads as "mild disagreement" and misses this completely. □

Exercise 4.9. Add a fifth organ P attached only to S, with $x_{\mathsf{P}} = x_{\mathsf{S}} = (1, 0)$: a perfectly aligned new organ contributing no disagreement. Recompute B, ρ, α and $\tilde{\rho}$. Which changed, and does the change match the dilution claim in item (i) above?

4.5 The dead dynamic range

Claims and non-claims. [REFUTED] (measured 2026-07-28). It was claimed that the compressed dynamic range of ρ is caused by encoder anisotropy acting through α. The claim does not survive. The deployed encoder returns unit-normalised vectors, for which

$$\alpha = \frac{(n-1)(1-\bar{c})}{n}, \tag{4.5}$$

with $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ and $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced *jointly* by α, by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$. On a seven-organ complex the measurement was $\tilde{\rho} \approx 0.47$ inside a window $[0.146, 0.854]$: the mode index has healthy dynamic range that ρ was multiplying away.

Exercise 4.10. Derive (4.5). Assume $\|x_v\| = 1$ for all v, take x_0 to be the constant assignment at the mean, and expand $\|x_\perp\|^2 = \sum_v \|x_v - \bar{x}\|^2$ in terms of pairwise inner products. The identity is three lines, and it is why the refutation is airtight rather than merely suggestive.

4.6 The precision-weighted generalisation

Construction 4.11 (Weighted discord and normaliser). **[IMPLEMENTED, UNPROVED]** Give each edge a *precision* $\pi_e > 0$ and set

$$\omega_\pi(x) = \sum_{e=\{u,v\}} \pi_e \left\| \mathcal{F}_{v \trianglelefteq e}(x_v) - \mathcal{F}_{u \trianglelefteq e}(x_u) \right\|^2, \tag{4.6}$$

the Dirichlet form of $L_\pi = \delta^\top P \delta$ with $P = \operatorname{diag}(\pi_e)$. Define the precision-weighted degree and the weighted normaliser

$$d_\pi(v) = \sum_{e \ni v} \pi_e, \qquad B_\pi = \max_{e=\{u,v\}} \big(d_\pi(u) + d_\pi(v)\big),$$

and replace B by B_π throughout.

Proposition 4.12 (The weighted bound). **[PROVED]** *In the identity-restriction regime,* $\lambda_{\max}(L_\pi) \leq B_\pi$.

Proof. For rational weights, the weighted graph Laplacian is the unweighted Laplacian of the multigraph in which edge e is replaced by π_e parallel copies; Theorem 4.1 applied there gives the bound with degrees replaced by weighted degrees. The general case follows because both sides are continuous in the weights and the rationals are dense. $\square$

> **Claims and non-claims.** This is where organ-level confidence legitimately enters the architecture. It does *not* enter as a stalk variance: the dimensional analysis of Chapter 5 shows a confidence-derived stalk floor is d-extensive and swamps everything else at $d = 384$. As an edge precision, entering as $s_e = -\ln c_e/d$, its trace contribution is $O(1)$ and it is dimensionally harmless. The same number is load-bearing in one place and destructive in the other, and the entire difference is where in the sheaf it is attached.

Part III

Semantic Geometry

Chapter 5

Semantic Geometry

Part I built the shape and Part II measured disagreement across it. This part supplies the geometry *inside* an organ: what a concept is, how many concepts collapse into the single vector a stalk holds, and how far apart two uncertain concepts are.

5.1 A concept is a Gaussian

> **Intuition.** A concept is not a point. It is a point plus an admission of how uncertain that point is, and — more subtly — of the *directions* in which it is uncertain. A concept may be sharply located along one axis of meaning and vague along another. A Gaussian with a structured covariance is the smallest object that records all three.

Definition 5.1 (Concept)**.** A *concept* is a Gaussian $\mathcal{N}(\mu_i, \Sigma_i)$ on $\mathbb{R}^d$ with low-rank-plus-isotropic covariance

$$\Sigma_i = U_i U_i^\top + D_i I_d, \qquad U_i \in \mathbb{R}^{d \times k_i},\ D_i > 0. \tag{5.1}$$

Here μ_i is the embedding, $U_i U_i^\top$ is anisotropic learned spread, and D_i is an isotropic noise floor — the epistemic humility of the concept.

The rank k_i is kept small (in practice $k_i \leq 8$). This is not a performance compromise: it is what makes every computation below reduce to a $k \times k$ problem rather than a $d \times d$ one, and it is the reason the geometry is affordable at all.

5.2 The coarse-graining π_v

> **Intuition.** An organ holds many concepts but exposes one vector to the coarse complex. Which vector? Not the plain average: a near-certain concept should dominate a vague one. The right notion of weight is *precision* — inverse variance — and the resulting formula is exactly how independent noisy measurements of a single quantity are optimally combined.

Construction 5.2 (The coarse-graining map)**.** For an organ v with active concepts $\{\mathcal{N}(\mu_i, \Sigma_i)\}_{i=1}^n$, the coarse stalk value is

$$\boxed{\pi_v \;=\; \Big(\sum_{i=1}^n \Sigma_i^{-1}\Big)^{-1} \Big(\sum_{i=1}^n \Sigma_i^{-1} \mu_i\Big)} \tag{5.2}$$

If v has no active concept then $\pi_v = \perp$, and v is excluded from the computation of ρ rather than contributing a zero vector.

Proposition 5.3 (π_v is the maximum-likelihood fusion). **[PROVED]** *Suppose $\theta \in \mathbb{R}^d$ is a latent vector and each concept is an independent observation $\mu_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \Sigma_i)$. Then the maximum-likelihood estimate of θ is (5.2).*

Proof. The log-likelihood is $\ell(\theta) = -\frac{1}{2}\sum_i(\mu_i - \theta)^\top \Sigma_i^{-1}(\mu_i - \theta) + \text{const}$. Its gradient is $\nabla_\theta \ell = \sum_i \Sigma_i^{-1}(\mu_i - \theta)$, and setting it to zero gives $\left(\sum_i \Sigma_i^{-1}\right)\theta = \sum_i \Sigma_i^{-1}\mu_i$. The matrix $\sum_i \Sigma_i^{-1}$ is positive definite (each Σ_i^{-1} is, since $D_i > 0$), hence invertible, and ℓ is strictly concave, so the stationary point is the unique maximum. □

Remark 5.4 (Three properties worth naming). The fusion (5.2) is *order-independent* (both sums are commutative), *non-destructive* (every concept contributes to both the precision sum and the mean; none is discarded), and *n-ary* rather than iterated-pairwise. These are not incidental: an order-dependent or lossy summary would make the coarse measurement depend on the sequence in which an organ happened to acquire its concepts.

Two computational paths, both exact

- **Isotropic fast path.** When all U_i are empty, $\Sigma_i^{-1} = D_i^{-1}I$ and (5.2) collapses to the scalar precision-weighted average
$$\pi_v = \frac{\sum_i D_i^{-1}\mu_i}{\sum_i D_i^{-1}}, \tag{5.3}$$
computable in $O(nd)$ with no matrix inversion.
- **General path.** With $U_i \neq \varnothing$, invert each Σ_i by the Sherman–Morrison–Woodbury identity
$$\Sigma_i^{-1} = D_i^{-1}I - D_i^{-1}U_i\left(I_{k_i} + D_i^{-1}U_i^\top U_i\right)^{-1}U_i^\top D_i^{-1}, \tag{5.4}$$
accumulate into a dense information matrix, and solve once, at $O(ndk + d^3)$.

Lemma 5.5 (Woodbury for this covariance). **[PROVED]** (5.4) *holds for $\Sigma = UU^\top + DI$ with $D > 0$.*

Proof. The Woodbury identity states $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$. Take $A = DI$, $C = I_k$, $V = U^\top$. Then $A^{-1} = D^{-1}I$ and $C^{-1} + VA^{-1}U = I_k + D^{-1}U^\top U$, which is positive definite and hence invertible. Substituting gives (5.4). The cost is dominated by the $k \times k$ inverse, not by anything of size d. □

Remark 5.6 (The floor is not optional). A lower bound $D_i \geq 10^{-6}$ is imposed. Without it, a concept reported as near-certain produces $1/D_i \approx 10^{16}$ and the information matrix loses all contribution from every other concept to rounding. The floor pins π_v strongly but finitely. Numerically the general path was checked against a dense $\left(\sum_i \Sigma_i^{-1}\right)^{-1}$ solve and agreed to 2.2×10^{-16}; equal precisions reduce to the plain mean, and a 9:1 precision ratio pulls π_v to the confident concept, as required.

Remark 5.7 (This map is the two-level glue). After execution, each organ's coarse stalk is *overwritten* by its own π_v. That is what makes the coarse ρ and the fine geometry one measurement at two resolutions rather than two unrelated numbers. It also means π_v is lossy in a load-bearing way: whatever the fine complex knew that does not survive (5.2) is invisible to every coarse diagnostic.

5.3 Distance between uncertain concepts

> **Intuition.** How far apart are two concepts? Comparing centres alone throws away the spreads. The 2-Wasserstein — optimal-transport — distance between Gaussians accounts for both, and is a genuine metric rather than a heuristic score.

Definition 5.8 (2-Wasserstein between Gaussians)**.**

$$W_2^2\big(\mathcal{N}(\mu_1,\Sigma_1),\mathcal{N}(\mu_2,\Sigma_2)\big) = \|\mu_1-\mu_2\|^2 + \operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 - 2\operatorname{Tr}\Big[\big(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\big)^{1/2}\Big]. \tag{5.5}$$

The covariance part alone is the squared Bures–Wasserstein distance d^2_{BW} on $\mathrm{SPD}(d)$.

Proposition 5.9 (Isotropic reduction to $O(k^3)$)**. [PROVED]** *If $\Sigma_1 = D_1 I$ is isotropic and $\Sigma_2 = U_2U_2^\top + D_2 I$ with $U_2 \in \mathbb{R}^{d\times k}$, then, writing $\{\sigma_j^2\}_{j=1}^k$ for the eigenvalues of the $k\times k$ matrix $U_2^\top U_2$,*

$$W_2^2 = \|\mu_1-\mu_2\|^2 + \big(\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2\big) - 2\sqrt{D_1}\Big[\sum_{j=1}^k \sqrt{\sigma_j^2 + D_2} + (d-k)\sqrt{D_2}\Big]. \tag{5.6}$$

Proof. Since $\Sigma_1^{1/2} = \sqrt{D_1}I$, we get $\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2} = D_1\Sigma_2$ and hence $\big(D_1\Sigma_2\big)^{1/2} = \sqrt{D_1}\,\Sigma_2^{1/2}$, so the cross term is $2\sqrt{D_1}\operatorname{Tr}\Sigma_2^{1/2}$. The eigenvalues of $\Sigma_2 = U_2U_2^\top + D_2 I$ are $\sigma_j^2 + D_2$ on $\operatorname{im} U_2$ and D_2 with multiplicity $d-k$ on $\ker U_2^\top$, because the nonzero eigenvalues of $U_2U_2^\top$ coincide with those of $U_2^\top U_2$. Summing $\sqrt{\lambda}$ over the spectrum gives (5.6). Only a $k\times k$ eigendecomposition is needed. ☐

This is exact for isotropic Σ_1, not a diagonal approximation.

5.4 The two terms are not commensurable

Construction 5.10 (Reporting the split)**.** W_2^2 is a sum of two quantities measuring different things, and they are returned separately rather than added:

$$W_2^2 = \underbrace{\|\mu_1-\mu_2\|^2}_{\text{semantic: } \textit{what} \text{ they say}} + \underbrace{\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 - 2\operatorname{Tr}\big[(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2})^{1/2}\big]}_{\text{epistemic: } \textit{how sure} \text{ they are}}. \tag{5.7}$$

Proposition 5.11 (Degeneration in the isotropic regime)**. [PROVED]** *If both concepts are isotropic, $\Sigma_i = D_i I$ with U_i empty, then*

$$W_2^2 = \|\mu_1-\mu_2\|^2 + d\big(\sqrt{D_1} - \sqrt{D_2}\big)^2. \tag{5.8}$$

In particular $D_1 = D_2$ implies $W_2^2 = \|\mu_1-\mu_2\|^2$ exactly.

Proof. Set $k=0$ in Proposition 5.9. Then $\operatorname{Tr}\Sigma_1 + \operatorname{Tr}\Sigma_2 = d(D_1+D_2)$ and the cross term is $2\sqrt{D_1}\cdot d\sqrt{D_2}$, so the covariance part equals $d\big(D_1 + D_2 - 2\sqrt{D_1D_2}\big) = d(\sqrt{D_1}-\sqrt{D_2})^2$. ☐

Claims and non-claims. Proposition 5.11 recorded two uncomfortable facts. Both were true when written; both were subsequently repaired. The diagnosis is retained because it is what forced the repair and because the structural lesson survives it.

(i) The machinery was inert. Every grown concept carried the same prior $D = 1.0$ with empty U, so $D_1 = D_2$ and W_2^2 was *exactly* squared Euclidean distance — an optimal-transport apparatus with a Wasserstein-shaped proof attached and nothing to do. After the repair, D is the shrinkage floor of Remark 5.15 and varies with effective sample size and with rank, so $D_1 \neq D_2$ in general. On two stalks with equal means and ranks 0 and 1, the epistemic term moves from 0 to 0.206: the geometry is live.

(ii) It would have become actively harmful the moment calibration was fixed. The semantic term is *d-intensive* — bounded by 4 for unit-normalised embeddings, whatever d is — while the epistemic term is *d-extensive*. At $d = 384$, a prior $D_1 = 1.0$ against a confident concept with $D_2 = -\ln 0.95$ gave an epistemic term of roughly 230 against a semantic term of 2: about 115 to 1, driven entirely by a self-reported confidence that the architecture elsewhere declines to trust. Merge, split and consolidation would then have been governed by hallucinated confidences. This is a latent defect that switches on precisely when the system is *improved*, which is the worst kind.

The lesson survives both fixes and is why the terms are reported apart: summing a d-intensive and a d-extensive quantity and thresholding the sum is a dimensional error, not a modelling preference.

Exercise 5.12. Verify the 115:1 figure. Take $d = 384$, $D_1 = 1.0$, $D_2 = -\ln 0.95$, both isotropic, and evaluate (5.8). Then repeat with the shrinkage floor $D = O(1/d)$ of Remark 5.15 and confirm the epistemic term drops to order 1. The point of doing this by hand is that the failure is visible in the exponents, before any numbers are substituted.

5.5 Fusion and its entropy

Definition 5.13 (Precision-weighted fusion)**.** The fusion of two concepts is the information-form product

$$\Lambda_i = \Sigma_i^{-1}, \qquad \Sigma_\star = (\Lambda_1 + \Lambda_2)^{-1}, \qquad \mu_\star = \Sigma_\star(\Lambda_1\mu_1 + \Lambda_2\mu_2), \tag{5.9}$$

which is (5.2) for $n = 2$. Full covariances are fused via Woodbury (5.4), never element-wise.

Proposition 5.14 (Fused entropy by the matrix determinant lemma)**.** **[PROVED]** *The differential entropy $h = \frac{1}{2}\ln\big((2\pi e)^d \det \Sigma_\star\big)$ is computable from a $k\times k$ determinant: for $\Sigma = UU^\top + DI$,*

$$\det \Sigma = D^d \det\big(I_k + D^{-1}U^\top U\big). \tag{5.10}$$

Proof. The matrix determinant lemma gives $\det(A + UV^\top) = \det A \cdot \det(I + V^\top A^{-1}U)$. With $A = DI$ and $V = U$ we have $\det A = D^d$ and $V^\top A^{-1}U = D^{-1}U^\top U$. □

Remark 5.15 (Where confidence actually goes). **[IMPLEMENTED, UNPROVED]** An earlier design set the noise floor from the generating model's self-reported confidence via $D = -\ln c$, with a named prior $D = 1.0$ when c was unavailable. Both were removed. The floor is now a shared shrinkage floor

$$D = \varepsilon + \frac{\kappa}{d\,(n_{\text{eff}} + \kappa)} = O(1/d),$$

independent of any reported confidence. The scaling is the whole point: D feeds a d-extensive Bures term, so a floor of order 1 contributes a trace of order d and swamps a semantic term bounded by 4. At $d = 384$ the two retired forms give isotropic traces of about 19.7 and 384 respectively; the shrinkage floor gives 0.88.

Confidence is *relocated, not discarded*: its home is the edge precision of Construction 4.11, entering as $s_e = -\ln c_e/d$ so that its trace contribution is $O(1)$. There it is dimensionally harmless and load-bearing; as an isotropic stalk variance it was neither. Against the deployed model, token log-probabilities are unavailable, so c is genuinely unknown and the engine warns once rather than fabricating a value. This is why ρ, and not self-reported confidence, is the primary signal.

> **Claims and non-claims.** The pattern in this chapter is worth extracting, because it recurs. Each of the three defects — inert geometry, dimensional mismatch, fabricated confidence — was invisible to every test that checked whether the code *ran*, and visible immediately to a check of whether the quantities being combined had the same units. Dimensional analysis is the cheapest correctness tool available here and it was applied late.

Part IV

Cohomology and Diagnostics

Chapter 6

Cohomology and the Diagnostic Hierarchy

6.1 Two pathologies that must not be confused

Definition 6.1 (Sheaf cohomology in low degrees)**.** With coboundaries $\delta^k : C^k(\mathcal{C};\mathcal{F}) \to C^{k+1}(\mathcal{C};\mathcal{F})$,

$$H^0(\mathcal{C};\mathcal{F}) = \ker \delta^0 \quad \text{(global sections)}, \qquad H^1(\mathcal{C};\mathcal{F}) = \ker \delta^1 / \operatorname{im} \delta^0 \quad \text{(gluing failure)}.$$

> **Intuition.** A *structural hole* — $b_1 \neq 0$, a Betti number of the bare shape — means a *binding is missing*: no theorem yet ties some loop together. A *gluing failure* — $H^1(\mathcal{C};\mathcal{F}) \neq 0$ of the sheaf — means the data are *contradictory*: the organs cannot be reconciled. The first calls for growth; the second calls for repair. Treating a contradiction as a hole makes you add structure when you should resolve a conflict, and treating a hole as a contradiction makes you argue when you should build.

Remark 6.2 (b_1 is earned here)*.* b_1 is a defensible diagnostic *only* because simplices were defined to mean genuine n-ary binding (Remark 1.3). A filled 2-simplex really asserts a joint relation, so an unfilled 1-cycle really is a missing joint relation. Most applications of topological data analysis to knowledge graphs skip this justification and compute b_1 of a complex whose 2-cells mean nothing in particular, at which point the number is an artefact of the construction rather than a fact about the knowledge.

6.2 The hierarchy is vacuous as currently fed

This is the most important negative result in the book, and it is worth stating as a proposition rather than as a caveat.

Proposition 6.3 (Exactness of the emitted cochain)**.** **[PROVED]** *Suppose the* 1*-cochain submitted for cohomological analysis is obtained as $y = \delta^0 x$ for some global state $x \in C^0(\mathcal{C};\mathcal{F})$. Then $[y] = 0$ in $H^1(\mathcal{C};\mathcal{F})$, for every x, every sheaf $\mathcal{F}$, and every choice of restriction maps.*

Proof. $H^1(\mathcal{C};\mathcal{F}) = \ker \delta^1 / \operatorname{im} \delta^0$ and $y \in \operatorname{im} \delta^0$ by construction, so y represents the zero class. (That $y \in \ker \delta^1$ is automatic from $\delta^1 \delta^0 = 0$, but is not even needed: membership in the image is already the conclusion.) □

Claims and non-claims. **[REFUTED]** The consequence is not subtle. The engine computes its 1-cochain by applying δ to the current state — that is what ω is built from. So the class it then measures in $H^1(\mathcal{C};\mathcal{F})$ is identically zero, and the growth mechanism that was supposed to be addressed by a nonzero class *has no address*. Every reported $H^1(\mathcal{C};\mathcal{F})$ diagnostic in that pipeline was measuring the definition of a coboundary, not the state of the system.

The repair requires a source of 1-cochains that is *not* of the form δx. The natural candidate is pairwise judgements formed independently: an organ pair that renders a verdict on their shared edge without either of them consulting a global state. That is new engineering — organs that emit pairwise judgements, storage for 2-simplices, and an actual δ^1 — and it does not exist yet.

One proposed repair in the sources is recorded by the audit as *half circular*, and the circularity must be located precisely before it is adopted. See Open Problem 17.2.

Remark 6.4 (The negative result is publishable)*.* Proposition 6.3 is not merely an internal correction. It identifies a failure mode that any sheaf-based diagnostic over a derived cochain will share, and the derivation is three lines. It belongs in the paper on cohomological obstruction as a stated negative result, not in a footnote.

6.3 The Hodge decomposition

Theorem 6.5 (Hodge decomposition for cellular sheaves)**.** **[PROVED]** *Let* $L_1 = \delta^0(\delta^0)^\top + (\delta^1)^\top\delta^1$ *be the degree-one Laplacian. Then*

$$C^1(\mathcal{C};\mathcal{F}) = \operatorname{im}\delta^0 \;\oplus\; \ker L_1 \;\oplus\; \operatorname{im}(\delta^1)^\top, \tag{6.1}$$

an orthogonal decomposition, and $\ker L_1 \cong H^1(\mathcal{C};\mathcal{F})$*.*

Proof. Orthogonality of $\operatorname{im}\delta^0$ and $\operatorname{im}(\delta^1)^\top$ follows from $\delta^1\delta^0 = 0$: for $y = \delta^0 a$ and $z = (\delta^1)^\top b$, $\langle y, z\rangle = \left\langle \delta^0 a, (\delta^1)^\top b\right\rangle = \left\langle \delta^1\delta^0 a, b\right\rangle = 0$. For $w \in \ker L_1$ we have $0 = \langle L_1 w, w\rangle = \left\|(\delta^0)^\top w\right\|^2 + \left\|\delta^1 w\right\|^2$, so $w \in \ker(\delta^0)^\top \cap \ker\delta^1$, which is the orthogonal complement of $\operatorname{im}\delta^0 \oplus \operatorname{im}(\delta^1)^\top$. Counting dimensions gives (6.1). Finally each class in $H^1(\mathcal{C};\mathcal{F}) = \ker\delta^1/\operatorname{im}\delta^0$ has a unique representative orthogonal to $\operatorname{im}\delta^0$, and that representative lies in $\ker L_1$. □

Intuition. The decomposition sorts every 1-cochain into three kinds of thing: the part that is the disagreement of some global assignment ($\operatorname{im}\delta^0$), the part that is a genuine irreducible contradiction ($\ker L_1$, the harmonic part), and the part that is a local circulation with no bearing on gluing. Proposition 6.3 says that as currently fed, the engine only ever produces cochains of the first kind.

Worked example. On the running example K_4 with the trivial sheaf ($\mathcal{F}(\sigma) = \mathbb{R}$): $\dim C^0(\mathcal{C};\mathcal{F}) = 4$, $\dim C^1(\mathcal{C};\mathcal{F}) = 4$ and the complex has one filled triangle so $\dim C^2(\mathcal{C};\mathcal{F}) = 1$. Since the 1-skeleton is connected, $b_0 = 1$ and $\operatorname{rank}\delta^0 = 4 - 1 = 3$. The only cycle in the 1-skeleton is the triangle $\{\mathsf{S},\mathsf{M},\mathsf{R}\}$, and it is filled, so $b_1 = 0$. Consistently: $\dim\ker\delta^1 = \dim C^1(\mathcal{C};\mathcal{F}) - \operatorname{rank}\delta^1 = 4 - 1 = 3 = \operatorname{rank}\delta^0$, so $H^1(\mathcal{C};\mathcal{F}) = 0$.

Now delete the 2-simplex, leaving the same 1-skeleton unfilled. Then $C^2(\mathcal{C};\mathcal{F}) = 0$,

$\ker \delta^1 = C^1(\mathcal{C}; \mathcal{F})$ has dimension 4, and $H^1(\mathcal{C}; \mathcal{F}) \cong \mathbb{R}^{4-3} = \mathbb{R}$, matching $b_1 = 1$: one unfilled loop, one class. This is the sense in which coning (Chapter 12) kills a class — it adds precisely the 2-cells that make the cycle a boundary. □

6.4 The uncertainty stack

Intuition. Coherence is not correctness. Five instruments, ordered by cost, answer five different questions, and they are not interchangeable. The most common error in systems of this kind is to let the cheapest instrument speak for all five.

Instrument	**Question**	**Cost**	**When**
ρ	Are my parts agreeing?	effectively free	every tick
$\hat{\rho}$ vs ρ	Did I *know* whether I would agree?	free (k-NN)	once history exists
$b_0,\ b_1$	What is structurally missing?	medium	on conflict
perturbation	Am I stable to paraphrase?	k model calls	high stakes only
VerifyOp	Am I actually *right*?	medium	promotion gate

Table 6.1: The five instruments and their distinct questions.

Remark 6.6 (Perturbation is a variance, not a derivative). Stability is measured as paraphrase-perturbation variance: run k semantically equivalent paraphrases and measure the spread of outcomes. This replaces an earlier proposal to compute a Fréchet derivative $\|\partial F/\partial x\|$ of the whole pipeline. That derivative does not exist: F factors through discrete token sampling, a DAG generator, and database lookups, none of which is differentiable. Earlier drafts also posited an infinite-dimensional Banach state space for the same purpose; it was removed for the same reason. The replacement costs k model calls and must be rationed.

Remark 6.7 (Expected precision and the feedback edge). $\hat{\rho}$ is a similarity-weighted k-nearest-neighbour prediction of ρ from a log of past (query embedding, ρ) pairs. The *signed* gap $\hat{\rho} - \rho$ separates three states that a single number cannot: *competence* (predicted agreement, observed agreement), *known ignorance* (predicted disagreement, observed disagreement — "confidently, I do not know this"), and *hallucination risk* (predicted agreement, observed disagreement). A VerifyOp failure updates the store, so self-knowledge improves with use. This is the only instrument in the stack whose accuracy is supposed to increase over time, and it is therefore the one that carries the architectural bet of Chapter 16.

Part V

Dynamics

Chapter 7

Growth Operators and the Algebra $\mathfrak{D}$

> **Intuition.** Learning *is* change of shape. The complex binds organs that work together, prunes idle ones, and — when a procedure repeatedly succeeds — promotes it to a first-class operator. Four axes of growth, one of which grows the very set of moves the system can make.

7.1 The structural operators

Definition 7.1 (Base moves)**.** The base moves on $\mathcal{C}$ are the typed operators

$$\{\mathsf{bind},\ \mathsf{collapse},\ \mathsf{split},\ \mathsf{merge}\},$$

each logged to an immutable mutation history.

Remark 7.2 ("Pachner move" was discarded, correctly)*.* Earlier drafts called these Pachner moves. A knowledge complex is not a manifold, so the invariance guarantees that make Pachner moves useful — that any two triangulations of a PL manifold are connected by a finite sequence of them — simply do not transfer. Borrowing the name would have imported a theorem that does not apply.

Construction 7.3 (The four growth axes)**.** Evolution proceeds along

$$\underbrace{\delta\mathcal{C}_{\text{fine}}}_{\text{concepts}},\ \underbrace{\delta\mathcal{C}_{\text{coarse}}}_{\text{coalitions}},\ \underbrace{\delta\mathcal{F}}_{\text{stalk and restriction data}},\ \underbrace{\delta\mathfrak{D}}_{\text{new operators}}.$$

The fourth axis is the essential one: the operator algebra is time-dependent, $\mathfrak{D}(t) \to \mathfrak{D}(t+1)$.

Definition 7.4 (Hebbian coupling and decay)**.** For a coalition σ with weight $w(\sigma,t) \in [0,1]$: co-activation increments w; idle steps apply $w(\sigma,t+1) = (1-\gamma)w(\sigma,t)$ for a decay rate $\gamma \in (0,1)$; a coalition falling below a bind threshold *collapses.* A pair past the bind threshold is a 1-simplex of K.

Proposition 7.5 (Idle coalitions collapse in finite time)**.** **[PROVED]** *If σ is idle from time t_0 and the bind threshold is $\theta > 0$, then σ collapses at the first $t \geq t_0 + \log_{1/(1-\gamma)}\big(w(\sigma,t_0)/\theta\big)$.*

Proof. Iterating the decay gives $w(\sigma, t_0 + m) = (1-\gamma)^m w(\sigma, t_0)$, which falls below θ as soon as $m \ln(1-\gamma) < \ln\big(\theta / w(\sigma, t_0)\big)$; since $\ln(1-\gamma) < 0$ this is $m > \log_{1/(1-\gamma)}(w(\sigma, t_0)/\theta)$. $\square$

Construction 7.6 (Consolidation and microneurons)**.** Over the weight filtration, persistent homology reads which coalitions and concept-cycles *survive*; survivors are promoted into permanent structure. A *microneuron* is a promoted sub-tree: when a composite DAG repeatedly resolves conflict *and* passes verification, it is named and thereafter emitted by the planner as a single atomic generator of $\mathfrak{D}$. This is $\delta\mathfrak{D}$ made concrete. The promotion gate is VerifyOp (Chapter 14): a composite becomes a generator only if it survives the external oracle.

> **Claims and non-claims.** The requirement that promotion pass VerifyOp is doing real work. Without it, "repeatedly resolved conflict" promotes whatever procedure most reliably makes the organs agree — which, in a system whose only cheap signal is agreement, selects for procedures that manufacture consensus rather than truth. Coherence is a control signal, not a fitness function, and the gate is what keeps that distinction enforced.

Chapter 8

The Fine Dynamics

> **Intuition.** Inside an organ, reasoning is a *flow*. New material perturbs the internal state and a gradient descent pulls it back toward internal coherence, subject to hard constraints (axioms that must hold) and soft constraints (inferred facts, bendable).

Construction 8.1 (Fine state and deductive flow)**.** The fine state of an organ is a configuration $x \in (\mathbb{R}^d)^m$ over its m active concepts, with a scalar *conflict functional* $\Phi(x) \geq 0$ measuring internal inconsistency — the fine-level analogue of ω. Reasoning integrates

$$\dot{x} = -\nabla\Phi(x) \tag{8.1}$$

by a fixed-step fourth-order Runge–Kutta scheme. The reported δ*-strain* of a step is the achieved change $\Delta\Phi$. Two constraint types shape Φ: *rigid* constraints (user-stated axioms) act as hard orthogonal projections onto an affine subspace via a projector P; *fluid* constraints (inferred facts) act as soft quadratic penalties.

Proposition 8.2 (The flow is a descent)**. [PROVED]** *Along any solution of* (8.1), $\frac{d}{dt}\Phi(x(t)) = -\left\|\nabla\Phi(x(t))\right\|^2 \leq 0$*, with equality exactly at critical points. If* Φ *is bounded below — which it is, by* $\Phi \geq 0$ *— then* $\Phi(x(t))$ *converges and* $\nabla\Phi(x(t)) \to 0$ *along a subsequence.*

Proof. Chain rule: $\frac{d}{dt}\Phi = \langle\nabla\Phi, \dot{x}\rangle = -\langle\nabla\Phi, \nabla\Phi\rangle$. Monotone and bounded below implies convergent; integrating $\int_0^\infty \left\|\nabla\Phi\right\|^2 dt = \Phi(x(0)) - \lim\Phi < \infty$ forces $\liminf\left\|\nabla\Phi\right\| = 0$. □

Remark 8.3 (Observed sign behaviour)*.* On a live run, injecting a new concept raised Φ by a structural jump (δ-strain $= +0.202$) and the subsequent flow pulled it back down (δ-strain $= -0.081$): a structural expansion followed by a deductive relaxation, which is the intended dynamics. Note that the expansion is *not* a descent step — it is an exogenous change of the functional itself, and only the relaxation is governed by Proposition 8.2.

> **Claims and non-claims.** The closed form of Φ is an implementation-level choice — a penalised quadratic on the fine complex — and is deliberately not over-specified here. What is mathematically committed is (8.1) (a gradient flow on a nonnegative conflict functional) and the rigid/fluid split as hard projection versus soft penalty. Convergence to a *global* minimum is not claimed and would be false for a non-convex Φ.

Chapter 9

Scheduling: Antichains, and Why "Operad" Is Not Earned

> **Intuition.** A reasoning act is a small program: a directed acyclic graph of operators, some depending on others. Independent operators can run at once, and the scheduler slices the DAG into layers of mutually independent work.

Definition 9.1 (Antichain foliation)**.** A reasoning act is a DAG $T = (N, \prec)$ of operator nodes. The scheduler foliates T into the antichain layers of $\prec$: layer 0 is the set of minimal nodes, and layer $\ell + 1$ is the set of nodes all of whose predecessors lie in layers $\leq \ell$. Each layer is executed in parallel. Correctness follows because nodes in one antichain share no $\prec$-dependency. Node *support* sets index which fine-complex vertices an operator reads and writes, so slices with disjoint support are safe to co-execute.

Remark 9.2 (The intended structure is a trace monoid)*.* What Definition 9.1 describes is antichain layering of a DAG. An operad requires multi-input composition together with equivariance, associativity and unit axioms, none of which are stated or used anywhere. The structure the design is reaching for is a *trace monoid* $\mathbb{M}(\Sigma, I)$ in the sense of Mazurkiewicz: the free monoid on the operator alphabet modulo $ab = ba$ for pairs in an *independence relation* I, whose canonical Foata normal form is exactly the antichain layering the scheduler computes. That identification would be a *sharpening* rather than a downgrade, since it supplies normal forms, a Hilbert series, and a canonical form for deduplication. The terminology should be corrected regardless of whether the identification is completed.

> **Claims and non-claims. [REFUTED]** The trace-monoid identification **does not currently hold**, and the reason is three separate defects, measured 2026-07-29 against the deployed operator set.
>
> *(i) The relation is not symmetric.* In the scheduler, admitting *any* empty-support node — including a read-only one — marks the whole slice as a global mutation and blocks every later node. So 8 of the 15 operator pairs can share a slice in one order and not in the other. An independence relation is symmetric by definition; a non-symmetric relation defines no trace monoid, hence no Foata normal form and no Hilbert series. This is very likely an unintended scheduling behaviour and is worth fixing on its own merits.
>
> *(ii) The support sets are placeholders.* Every support accessor returns a hardcoded

literal — $\varnothing$, $\{0\}$ or $\{1\}$ — with three of the six operators returning the same $\{0\}$, and a comment standing in for an analysis of what the operator actually touches. Any invariant computed from these describes six constants, not the system.

(iii) The convention for $\varnothing$ is inverted. In the engine an empty support means GLOBAL: touches everything, must run isolated. Read as a set, $\varnothing$ is disjoint from everything and would mean independent of everything. The two readings give capacities differing by a factor of 1.8 ($\kappa = 3.00$ versus $\kappa = 5.45$, against $\kappa = 6$ for the free monoid), so the quantity is not well defined until the convention is fixed.

Under the reading the engine actually implements, the algebra is *nearly free* — about 9% below the free monoid — because three of six operators share one support. The order of repair is: fix the symmetry defect; derive real read/write sets, including the shared state the current model ignores entirely (the store, the obstruction counter Ω, the mutation log, and the coarse stalks that π_v overwrites after every execution); only then recompute I. Operators that all mutate Ω do not commute whatever their vertex supports say. Until that is done, no capacity, Koszulity, or resolution claim about $\mathfrak{D}$ should appear anywhere.

Chapter 10

The Lift–Project Adjunction

Intuition. Two worlds: free-form natural language, and structured reasoning DAGs. One map *lifts* language into a validated DAG; a partner map *projects* a DAG back into language. Meaning that carries no reasoning content collapses under the lift — it lands in the kernel.

Construction 10.1 (The boundary morphisms)**.** Let $\mathcal{L}$ send a natural-language request to a validated reasoning DAG, and $\mathcal{M}$ send a DAG to a natural-language rendering. The lift is guarded by a validator rejecting malformed DAGs at the boundary: duplicate identifiers, dangling children, cycles by depth-first colouring, absence of a terminal output node, and disconnection. The downstream engine therefore only ever receives well-formed programs.

Remark 10.2 (The kernel behaves as intended)*.* Semantically void input maps into $\ker \mathcal{L}$: a joke embedded in a mathematical prompt produced zero DAG nodes, exactly as a kernel element should. This is an observation on one input, not a theorem.

Claims and non-claims. $\mathcal{L} \dashv \mathcal{M}$ is the intended categorical *framing* — lift left adjoint to project — and it organises the design usefully. The unit and counit are not constructed here and the triangle identities are not verified, so "adjunction" should be read as a structuring principle plus two concrete morphisms with an observed kernel behaviour, and not as a proved theorem. Stating this is part of the honesty invariant rather than an apology for it.

Part VI

The Later Constructions

Chapter 11

The MDL Growth Law

The question this chapter answers: *when is a new concept worth inventing?* Earlier designs answered it with a threshold someone chose. This one derives a threshold instead.

11.1 Two-part description length

> **Intuition.** A memory is a compression of experience. Minimum description length makes that literal: the total cost of an explanation is what it takes to write down the model plus what it takes to write down the data *given* the model. Adding a concept always makes the first term worse. It is worth doing exactly when it makes the second term better by more.

Definition 11.1 (Two-part MDL)**.**

$$L_{\text{total}} = \underbrace{L(\text{model})}_{\text{the memory}} + \underbrace{L(\text{data} \mid \text{model})}_{\text{the experience}}.$$

The *model* is the complex $\mathcal{C}$ together with the sheaf $\mathcal{F}$ — what the system *is*. The *data* is the observed record: the sequence of assemblies and the succession relation read off it — what the system has *seen*.

Neither term alone is the criterion. Minimising the model term alone forbids all growth; minimising the data term alone is achieved by memorising everything. The criterion is the sum.

11.2 The model term

Proposition 11.2 (Cost of coning a cycle)**. [PROVED]** *Let γ be a 1-cycle of length k, and cone it with a new vertex w carrying a stalk of dimension d_w. Writing b for bits per real number, the added model cost is*

$$\Delta L(\textit{model}) = b\, d_w \Big(1 + \sum_{e\in\gamma} d_e\Big). \tag{11.1}$$

Proof. Three kinds of new object are described. The vertex with its stalk costs $d_w \cdot b$ bits. Each of the k restriction maps out of $\mathcal{F}(w)$ is a $d_e \times d_w$ matrix, costing $d_w d_e b$ bits, summed over $e \in \gamma$. The k new triangles cost *nothing*: they are determined by the cone construction, not described separately. Summing gives (11.1). □

Remark 11.3 (Two consequences, both load-bearing). First, the cost is *linear in* d_w. The dimension of the new stalk is the dominant free quantity, so whatever fixes d_w fixes most of the cost — which is what Section 12.6 is for. Second, *triangles are free.* They are implied by the cone, so the k new 2-cells cost nothing to describe. This is a real asymmetry, not a bookkeeping convenience, and it is why coning is cheap relative to what it buys.

11.3 The data term and the law

Proposition 11.4 (Data-term saving). **[PROVED]** *Let the cycle γ occur n times in the record. Without the concept, each traversal is coded as k separate transitions, each costing its full surprisal; with the concept it is a single named event. Then*

$$\Delta L(data \mid model) = -n\big(c_{old} - c_{new}\big), \tag{11.2}$$

where c_{old} is the cost of coding one traversal the long way and c_{new} the cost of naming it.

Theorem 11.5 (The growth law). **[PROVED]** *Attach the concept if and only if $\Delta L_{total} < 0$, that is*

$$\boxed{n \;>\; \frac{b\, d_w\big(1 + \sum_{e\in\gamma} d_e\big)}{c_{old} - c_{new}}} \tag{11.3}$$

Proof. Add (11.1) and (11.2) and require the sum negative; rearrange, using $c_{\text{old}} > c_{\text{new}}$. □

> **Intuition.** Read what this is: a *recurrence threshold that is computed, not chosen.* It is the ratio of what a concept costs to store against what it saves per use — the number of times a pattern must repeat before it is worth naming.

Three consequences, two of which were not asked for:

1. n is a number the engine can print, rather than a parameter a human sets.
2. **It orders the holes.** When several harmonic classes compete, fill them in decreasing $|\Delta L_{\text{total}}|$. This removes a decision that had been silently delegated to a human.
3. **It predicts non-attachment.** A one-off inconsistency is *noise to be tolerated*, not a concept to be invented. The earlier threshold-based law had no way to express this, which is exactly how it would have over-generated.

Chapter 12

The Coned Concept

This is the central construction of the later work: the stalk of a newly-grown concept, *derived* rather than chosen.

12.1 The topological cone

Construction 12.1 (Coning a cycle)**.** Let $\gamma = (v_1, e_1, v_2, \ldots, v_k, e_k, v_1)$ be a 1-cycle carrying harmonic mass, with $e_i = \{v_i, v_{i+1}\}$ (indices mod k). Attach a vertex w, edges $f_i = \{w, v_i\}$, and triangles $t_i = \{w, v_i, v_{i+1}\}$.

Proposition 12.2 (The cone kills the cycle)**.** **[PROVED]** *With $\partial t_i = e_i - f_{i+1} + f_i$,*

$$\partial\Big(\sum_i t_i\Big) = \sum_i e_i + \sum_i (f_i - f_{i+1}) = \gamma,$$

since the f terms telescope to zero. Hence γ becomes a boundary.

12.2 The stalks, chosen minimally

Construction 12.3 (New edge stalks)**.** Set

$$\mathcal{F}(f_i) := \mathcal{F}(v_i), \qquad \mathcal{F}_{v_i \trianglelefteq f_i} := \mathrm{Id}.$$

> **Claims and non-claims.** This is an *engineering choice*, and a deliberately empty one. Any distortion introduced here would be a free parameter, and whatever the construction then achieved could be attributed to a cleverly chosen edge stalk rather than to the concept. All the content is forced into $\mathcal{F}(w)$ and the maps out of it, which is where it can be reasoned about.

The only remaining unknowns are $\mathcal{F}(w)$ and the legs $p_i := \mathcal{F}_{w \trianglelefteq f_i} : \mathcal{F}(w) \to \mathcal{F}(v_i)$.

12.3 The cone condition is forced, not imposed

Proposition 12.4 (Derivation of the cone condition)**.** **[PROVED]** *Write $r_i^- := \mathcal{F}_{v_i \trianglelefteq e_i}$ and $r_i^+ := \mathcal{F}_{v_{i+1} \trianglelefteq e_i}$. For a 0-cochain x:*

(i) δx *vanishes on the new edges if and only if* $x_{v_i} = p_i(x_w)$ *for every* i*;*

(ii) given (i), δx *vanishes on the old edges for* every $x_w \in \mathcal{F}(w)$ *if and only if*

$$\boxed{r_i^+ \circ p_{i+1} = r_i^- \circ p_i \quad \textit{for all } i} \tag{12.1}$$

Proof. (i) On the new edge f_i, using Construction 12.3,

$$(\delta x)_{f_i} = \mathcal{F}_{w \trianglelefteq f_i} x_w - \mathcal{F}_{v_i \trianglelefteq f_i} x_{v_i} = p_i(x_w) - x_{v_i},$$

which vanishes exactly when $x_{v_i} = p_i(x_w)$.

(ii) Substituting $x_{v_i} = p_i(x_w)$ into the old edge e_i,

$$(\delta x)_{e_i} = r_i^+ p_{i+1}(x_w) - r_i^- p_i(x_w) = \big(r_i^+ p_{i+1} - r_i^- p_i\big)(x_w).$$

A linear map vanishes on all of $\mathcal{F}(w)$ if and only if it is the zero map, which is (12.1). □

> **Intuition.** Part (i) says something worth pausing on: **the value at w generates the section over the whole cycle**. That is what it means for a concept to explain a loop — the loop stops being k independent facts and becomes one.

Condition (12.1) is verbatim the definition of a *cone* over the diagram

$$D_\gamma : \quad \mathcal{F}(v_1) \xrightarrow{r_1^-} \mathcal{F}(e_1) \xleftarrow{r_1^+} \mathcal{F}(v_2) \xrightarrow{r_2^-} \mathcal{F}(e_2) \longleftarrow \cdots$$

with apex $\mathcal{F}(w)$ and legs p_i. The categorical cone condition was not imposed on the construction; it fell out of demanding that the topological cone do its job.

12.4 The construction

Construction 12.5 (The coned concept)**.**

$$\boxed{\mathcal{F}(w) := \varprojlim D_\gamma, \qquad p_i := \text{the limit's projections.}}$$

Proposition 12.6 (Concrete description)**. [PROVED]** *For finite-dimensional stalks,*

$$\varprojlim D_\gamma = \Big\{ (x_1, \ldots, x_k) \in \bigoplus_i \mathcal{F}(v_i) \;\Big|\; r_i^- x_i = r_i^+ x_{i+1} \ \forall i \Big\} \;\cong\; H^0\big(\gamma; \mathcal{F}|_\gamma\big).$$

Proof. The limit of a finite diagram of vector spaces is the subspace of the product on which all the diagram's maps agree, which is the displayed equaliser. Comparing with Definition 3.1 restricted to the subcomplex γ, the agreement conditions are exactly $\delta x = 0$, so the space is $\ker \delta|_\gamma = H^0(\gamma; \mathcal{F}|_\gamma)$. □

> **Intuition.** In words: the new concept *is* everything that was consistent around the loop but could not be glued. It does not summarise the cycle's members; it *is* the compatible part of them.

Three properties, each of which had been written down separately as a requirement:

Property	Why it matters
canonical — unique up to unique isomorphism	no hand-made choice enters the construction
path-dependent — depends on D_γ, i.e. on which cycle history produced	precisely the property whose absence caused an earlier formal-concept-analysis approach to be rejected
degenerate case	if $H^0(\gamma; \mathcal{F}\|_\gamma) = 0$ the limit is the zero space — but see Proposition 12.7

12.5 The refusal branch is unreachable

Proposition 12.7 (Harmonic mass implies sections). **[PROVED]** *On a k-cycle with vertex and edge stalks all of dimension n, and with no 2-cells,*

$$\dim H^1(\gamma; \mathcal{F}|_\gamma) = \dim H^0(\gamma; \mathcal{F}|_\gamma).$$

Consequently the cycle carries harmonic mass if and only if it has nonzero sections.

Proof. For a 1-dimensional complex the Euler characteristic satisfies $\chi = \dim C^0(\mathcal{C}; \mathcal{F}) - \dim C^1(\mathcal{C}; \mathcal{F}) = \dim H^0 - \dim H^1$. On a k-cycle with all stalks of dimension n there are k vertices and k edges, so $\dim C^0(\mathcal{C}; \mathcal{F}) = \dim C^1(\mathcal{C}; \mathcal{F}) = kn$ and $\chi = 0$, giving the equality. With no 2-cells the harmonic space *is* H^1. □

Claims and non-claims. An earlier draft claimed the degenerate case is meaningful — "nothing was consistent around that loop, so the construction refuses rather than inventing". True, and it never happens. By Proposition 12.7, $H^0 = 0$ implies no harmonic mass, which implies no growth address, which implies the growth law never fires on that cycle at all. Confirmed numerically: the degenerate case reports that the harmonic space is already empty before coning, so there is nothing to kill.

In general the gap is $\dim H^1 - \dim H^0 = k(n_e - n_v)$, so the branch *could* become reachable if edge stalks are ever given a different dimension from vertex stalks. They are not, currently.

12.6 The limit is a ceiling, not a floor

Claims and non-claims. [REFUTED] An earlier draft justified Construction 12.5 by asserting that the limit is "the smallest object admitting the required maps". That is backwards, and the error is worth keeping visible.

Proposition 12.8 (Universality bounds from above). **[PROVED]** *Every cone over D_γ factors uniquely through the limit. If a candidate apex $\mathcal{F}(w)'$ has jointly injective legs — i.e. carries no data that no restriction map ever sees — then that factoring map is injective, so*

$$\dim \mathcal{F}(w)' \;\leq\; \dim \varprojlim D_\gamma.$$

Proof. The universal property of the limit gives a unique $\phi : \mathcal{F}(w)' \to \varprojlim D_\gamma$ commuting with the legs. If $\phi(z) = 0$ then every leg $p_i'(z) = 0$; joint injectivity of the legs then forces $z = 0$, so ϕ is injective and the dimension inequality follows. □

So the limit is the *largest non-redundant* choice, not the smallest — a ceiling on what a new concept can usefully carry. The two determinations are therefore genuinely separate, and both are derived:

1. **Consistency fixes the shape.** $\mathcal{F}(w)$ must be a cone over D_γ; the limit is the universal one.
2. **MDL fixes the size.** By (11.1) the cost is linear in d_w, so the optimum is a *subspace* of the limit: keep the directions that pay for themselves, discard the rest.

Chapter 13

Retrieval: Rank, Do Not Threshold

> **Intuition.** Retrieval decides which stored concepts an organ sees. A threshold on distance answers "which concepts are close enough?", which sounds right and is not: it makes the number of results depend on how the embedding space happens to be scaled, and it silently returns nothing at all when the query lands in a sparse region.

Construction 13.1 (Bounded rank query)**.** Replace the ε-ball query with a bounded rank query returning the top k stored concepts, ranked by *cosine similarity* to the query mean, using a bounded max-heap of size k. Zero-norm vectors are dropped rather than scored, since a zero vector has no direction.

Proposition 13.2 (Why cosine and not squared distance)**.** **[PROVED]** *Expanding,*

$$\|\mu_q - \mu_i\|^2 = \|\mu_q\|^2 - 2\,\langle \mu_q, \mu_i \rangle + \|\mu_i\|^2 .$$

For a fixed query, $\|\mu_q\|^2$ is constant and drops out of any ranking. The term $\|\mu_i\|^2$ does not. *Hence squared distance reproduces the cosine ranking if and only if every stored mean is unit-norm.*

> **Claims and non-claims.** Stored means are *not* unit-norm. By Construction 5.2, π_v is a weighted centroid of unit vectors, so its norm lies below 1 and falls further the more spread the concepts it summarises. Ranking by squared distance would therefore apply a per-concept penalty *proportional to how broad a concept is*, pushing exactly the general concepts down the list for a reason unrelated to the query.
>
> There is also an evidential argument, which is decisive on its own: the measured recall figure against which the retrieval fix is accepted was obtained on a *cosine* ranking. Ranking by squared distance is a different ordering carrying an unmeasured bias, which would have made the acceptance criterion meaningless.

Remark 13.3 (Cost). Scoring needs only μ; the low-rank factor U, the floor D and the stored reasoning text are not needed unless the row enters the heap. Reading μ, scoring, and deserialising the rest only on entry drops the per-row cost from $O(dk^2 + k^3)$ to $O(d)$ for the large majority of rows.

Claims and non-claims. k is an *engineering choice* ($k = 24$), not a derived quantity. What changed is that it is now constrained on both sides — too small starves the organ, too large defeats the point of retrieval — rather than being an unexamined constant. Saying so is the difference between a parameter and a magic number.

Part VII

Verification and Control

Chapter 14

Verification: the External Oracle

Intuition. Coherence governs whether to keep exploring or stop and reconcile. Only an external checker touches *truth*. And "I could not check" must never be mistaken for "it is false".

Definition 14.1 (Three-valued verification)**.** VerifyOp calls a sandboxed symbolic oracle returning one of three verdicts, acting on an obstruction counter Ω:

exit	verdict	effect on state
0	Verified	obstruction cleared; state may crystallise
1	Refuted	Ω += 1000; forced into Resolve
2	Unverifiable	Ω *untouched*

Remark 14.2 (Absence of proof is not proof of absence)*.* An earlier version treated every nonzero exit as failure, conflating Refuted with Unverifiable. That would spike conflict over the oracle's own blind spots and lock the system into Resolve over perfectly good mathematics. A checker crash also returns Unverifiable. This three-valued logic is the promotion gate of Construction 7.6: a composite becomes a generator of $\mathfrak{D}$ only if it survives the oracle, and surviving means Verified, not merely "not refuted".

Claims and non-claims. The asymmetry in the table is deliberate and is the load-bearing design decision. A refutation moves Ω by a large fixed amount; an inability to check moves it not at all. The system is therefore allowed to be ignorant without being punished for it, and is punished immediately for being wrong. The alternative — treating unverifiable as mildly bad — produces a gradient toward only ever asserting things the oracle happens to understand, which is a narrower system, not a better one.

Chapter 15

The Controller

Definition 15.1 (Coherence-gated mode)**.** With threshold ε, the controller selects

$$\mathrm{mode}(x) = \begin{cases} \textsc{Resolve}, & \rho > \varepsilon, \\ \textsc{Explore}, & \rho \le \varepsilon, \\ \textsc{Unknown}, & \rho = \perp. \end{cases} \tag{15.1}$$

Remark 15.2 (Three modes, and an empirical ε)*.* The UNKNOWN branch is essential: when ρ is undefined the controller *refuses* to default to EXPLORE, because "no measurement" is not "all is well" (Remark 4.5). The threshold is calibrated rather than guessed: on real 384-dimensional embeddings, agreement gave $\rho \approx 0.00$, related-but-distinct ≈ 0.06, and off-topic ≈ 0.14, so $\varepsilon \approx 0.10$ is a principled starting gate.

Two cautions were recorded from live runs. First, ρ is *not* monotonic in the number of active organs: adding a well-aligned organ raises the normaliser faster than the discord, so ρ dilutes. Second, comparisons across differently sized complexes must be made with care, because B and $\|X\|_F^2$ both change.

Remark 15.3 (Superseded: gate on $\tilde{\rho}$, not ρ)*.* Both cautions are consequences of a single fact, isolated in Proposition 4.6: $\rho = \alpha \cdot \tilde{\rho}$, and *only α has the pathologies.* Organ-count dilution moves α and leaves $\tilde{\rho}$ invariant. Cross-complex comparison is unsafe for ρ, but $\tilde{\rho}$ is comparable across complexes once expressed as its position inside the window (4.4). The controller should therefore gate on $\tilde{\rho}$ — or on its window position — and report α alongside as the answer to a genuinely different question.

The typed reading is what the single number could not give:

- large α, small $\tilde{\rho}$: a clean two-camp split; the coalition wants restructuring (`split`, or a new `bind`);
- small α, large $\tilde{\rho}$: one organ is an outlier; a single edge wants repair.

> **Worked example.** Apply the controller to the running example. With $\varepsilon = 0.10$ and $\rho = 0.2$, mode (15.1) returns RESOLVE, and a diagnostic reading only ρ would report "mild disagreement, reconcile". The typed reading gives more: $\alpha = 0.5$ is large and $\tilde{\rho}$ sits at window position $\frac{1}{3}$, so by Remark 15.3 this is a two-camp split, and the correct remediation is `split` or a new `bind` — *not* repair of the worst edge. Note also that the worst-edge diagnostic was ambiguous here (the argmax tied), which is the same fact appearing in a

second instrument. □

Claims and non-claims. The controller is the one place where every earlier honesty decision becomes operational. $\perp$ propagates into a mode rather than being coerced to a number; the split of Proposition 4.6 changes which quantity is gated on; the window of Proposition 4.8 makes the threshold topology-relative. It is also where a mistake is most expensive, because a controller that mis-reads its instrument does not merely report wrongly — it acts.

Part VIII

Limits: What Is and Is Not Claimed

Chapter 16

Limits: What Is and Is Not Claimed

Intuition. The deepest thing the topology buys is not that it makes the system smart. It is that it makes the claim of augmentation *falsifiable*: memory, consistency and growth become operations on one object, with cheap provable guarantees and a live measure of whether augmentation is actually happening. A framework that cannot fail is not doing any work, and most of this chapter is about where this one can.

16.1 The bet and the refusal

Claims and non-claims. The bet (defensible). A persistent, self-checking, self-restructuring *organiser* wrapped around a fixed language model can, over time, produce better-grounded and more internally consistent research artifacts than the raw model, through measured cross-organ error correction and accumulation. The advantage is *architectural*, expressed as operations on the single object $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$.

The refusal. No simplex proves a theorem and no Laplacian integrates a differential form. The per-inference reasoning quality is bounded by the underlying model. The mathematics here does not raise that ceiling; it organises, measures and error-corrects *around* it.

The bet is about the *slope*, not the intercept. That makes it an empirical claim, and Open Problem 17.3 records that it has not been tested.

16.2 The scope is three papers, not one

Claims and non-claims. A single document containing the ρ splitting, the operator algebra, the coned concept, the cohomological obstruction and a spectral prediction on $\mathbb{CP}^3$ will produce something no referee can accept, because those claims have incompatible standards of evidence. The audit's recommended split is:

Paper A — a cellular-sheaf coherence measure for multi-agent orchestration. Content: $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$; ω and ρ; the $\alpha/\tilde{\rho}$ splitting with the window; weighted Anderson–Morley *with* the $\|\mathcal{F}\| \leq 1$ hypothesis stated; worst-edge localisation; the three-valued verification gate. Every claim is a theorem plus a measurement on a running system. This paper is nearly writable today.

Paper B — sheaf-cohomological obstruction as a diagnostic. Content: Proposition 6.3 — the derived cochain is exact, a negative result worth publishing on its own; the architecture needed to produce non-derived 1-cochains; the Hodge trichotomy; and the connection to contextuality in the Abramsky–Brandenburger sense, which is the thread that would make this a quantum-information contribution rather than a software paper. This paper *requires new engineering* and cannot be written before that lands.

Paper C — the spectral work. This is separate mathematics. It has nothing to do with TATIANA and should not be contaminated by it; the architecture's role is to be the tool that helped, mentioned in a methods note, not a co-author of the theorem.

16.3 What must not be claimed

Carrying forward the honesty discipline that is the best thing about the source corpus:

- **Not** "we prove the whole exceeds the sum". A structural novelty statement says a composite is *new*, not that it is true or useful. Verification remains the sole arbiter and runs first.
- **Not** "capacity measures learning". It is gameable by adding independent useless operators. Report it; never optimise it.
- **Not** "the system uses D-module theory". It uses Gaussians, which *are* the simplest holonomic modules — an exact identification that currently buys three invariants, all of them constant. The defensible claim is that the concept type admits a conservative extension to holonomic modules, feasible only because of the existing low-rank design.
- **Not** "Riemann–Hilbert applies". One leg is rigorous (cellular sheaves are constructible sheaves via exit paths); the other requires a commitment to an algebraic stratification that has not been made.
- **Not** "the threshold is derived". It is *normalised* (Proposition 4.8).
- **Not** "b_1 measures conceptual gaps" without Remark 6.2. The justification is the whole content of the claim.
- **Not** "the architecture is an operad". See Remark 9.2 and the three defects following it.

16.4 The findings, restated as boundaries

The audit's fourteen findings are not a list of bugs; each marks a place where a claim outruns its evidence. Restated as standing limits:

1. **The audit itself was against a stale snapshot.** Several findings were repaired before the audit was read. Any use of this list must be re-checked against the current branch — and that instruction applies to this book.
2. **Theorem 7.5 is stated more generally than proved.** The gap between hypothesis and proof has not been located precisely, and until it is, the theorem should be cited in its proved form only.
3. **The $H^1(\mathcal{C};\mathcal{F})$ repair is half circular.** The circularity must be identified before the repair is adopted (Open Problem 17.2).
4. **Proposition 12.4 is false in the deployed case.** A counterexample exists and must be reproduced in Appendix C before that entry is complete.

5. **The K_0 classes are trivial for every object the system produces.** An invariant that separates nothing is not a diagnostic (Open Problem 17.7).
6. **The independence relation is not symmetric**, its supports are placeholders, and its empty-set convention is inverted (Chapter 9).
7. **The restriction maps are the identity**, so the sheaf is a graph Laplacian in disguise (Open Problem 17.1).

Claims and non-claims. The honest summary of the whole corpus: the *measurement* layer is real, proved and running — ω, ρ, the splitting, the window, the three-valued gate, the coned concept and the growth law all survive scrutiny. The *expressive* layer is not yet built: while restrictions are the identity and 1-cochains are derived, the sheaf and its cohomology are carrying no information the graph did not already carry. That is not a fatal criticism; it is a precise statement of what the next piece of work has to be, and the fact that it can be stated this precisely is what the formalism bought.

Part IX

Open Problems within MOS

Chapter 17

Open Problems within MOS

These are stated as problems, not as directions, because each one has a definite answer that is currently unknown and whose resolution would change what the system can claim. They are ordered by how much else depends on them.

Open Problem 17.1 (Non-identity restriction maps). Find a procedure that learns restriction maps $\mathcal{F}_{v \trianglelefteq e}$ from organ data such that (i) the maps are not all the identity, (ii) functoriality (2.1) holds on every filled simplex, and (iii) the learning is closed-form or otherwise costs no gradient training. Closed-form Procrustes alignment satisfies (i) and (iii) but is not known to satisfy (ii). Everything expressive about the sheaf is downstream of this problem: while the maps are the identity, $\ker L$ is the diagonal and the sheaf is a graph Laplacian in disguise (Proposition 3.6).

Open Problem 17.2 (A non-vacuous $H^1(\mathcal{C};\mathcal{F})$). Under identity restrictions the cohomological hierarchy carries no information: the 1-cochain the engine emits is δx for a global state x, and a coboundary is exact by definition, so its class in $H^1(\mathcal{C};\mathcal{F})$ vanishes identically — for every state, every sheaf, and every choice of restriction map. Find a source of 1-cochains that is *not* of the form δx — pairwise judgements formed independently of any global state are the candidate — and determine whether the resulting classes are stable enough to serve as a diagnostic. The audit records that one proposed repair is half circular; the circularity must be identified precisely before a fix is adopted.

Open Problem 17.3 (Does the architecture actually compound?). The central bet is on the *slope*, not the intercept: that a self-checking, self-restructuring organiser improves with use even though its per-inference ceiling is fixed by the underlying language model. This is an empirical claim and it has not been tested. Design an experiment with a pre-registered metric, a control (the same model with no architecture), and a horizon long enough for accumulation to show, and state in advance what result would falsify the bet.

Open Problem 17.4 (The right normaliser). The Anderson–Morley bound B over-bounds $\lambda_{\max}$, which compresses ρ toward zero and contributes to its dead dynamic range. Determine whether a tighter, still eigensolver-free bound exists for the sheaf Laplacian with non-identity restrictions, or prove that no degree-based bound can be tight in this regime.

Open Problem 17.5 (Stability of the diagnostics under growth). The growth operators change the topology, and both the calibration window and the kernel projection are topology-dependent. Give a stability estimate: if `bind` or `collapse` changes the complex by one simplex, how much can $\tilde{\rho}$ move for a fixed state? Without such an estimate, a change in $\tilde{\rho}$ across a growth step cannot be attributed to the state rather than to the rewiring.

Open Problem 17.6 (Functoriality repair by hybridisation). When restriction maps are obtained as projections onto an interface, functoriality fails predictably. The canonical repair is to promote the interface to a cell of its own, which is what hybridisation methods do in the finite-element setting. Determine whether the same repair applies to the cognitive complex, and what it costs: does promoting every offending interface preserve the two-level stratification, or does it force a third stratum?

Open Problem 17.7 (Composition factors as memory atoms). A later development formalises memory atoms as Jordan–Hölder composition factors, with retrieval ordered by stratum, then integer content, then structure. Determine whether the composition series is canonical for the objects the system actually produces — the audit records that the associated K_0 classes are trivial for every such object, which if correct means the invariant distinguishes nothing.

Part X

Potentials for Quantum Implementation

Chapter 18

Potentials for Quantum Implementation

Claims and non-claims. **[PROPOSED]** Nothing in this part is implemented, and nothing in it is required by any result elsewhere in the book. It is included because the sheaf Laplacian is exactly the kind of object for which quantum linear algebra has a story, and because that story has a specific shape worth recording before it is attempted. Every claim here should be read as a research programme with an identified failure mode, not as a capability.

18.1 Why this object at all

The two quantities TATIANA computes on every tick are a Dirichlet energy $\omega(x) = x^\top Lx$ and a normalised version of it. Both are quadratic forms in a sparse, symmetric, positive semidefinite operator built from local data. This is the standard input to quantum linear-algebra routines, and the standard caution applies: the speedups are in the linear algebra, and they are destroyed by the cost of getting classical data in and scalar answers out.

18.2 The Dirac operator and its block-encoding

Definition 18.1 (Sheaf Dirac operator)**.** Let $d_{\mathcal{F}} = \bigoplus_k \delta^k$ be the total coboundary of the sheaf. The *sheaf Dirac operator* is

$$D_{\mathcal{F}} \ = \ d_{\mathcal{F}} + d_{\mathcal{F}}^\dagger,$$

a self-adjoint operator on $\bigoplus_k C^k(\mathcal{C}; \mathcal{F})$ satisfying $D_{\mathcal{F}}^2 = \Delta$, the total Hodge Laplacian. Its kernel is the space of harmonic cochains, which by Hodge theory is isomorphic to $\bigoplus_k H^k(\mathcal{C}; \mathcal{F})$.

The programme, stated as a pipeline:

1. **Block-encode $D_{\mathcal{F}}$.** Construct a unitary U whose top-left block is $D_{\mathcal{F}}/\gamma$ for a subnormalisation $\gamma \geq \|D_{\mathcal{F}}\|$. For a cellular sheaf with *matrix-valued* restriction maps this is the substantive step, and the subnormalisation γ scales with the heterogeneity of those matrices — which is precisely the quantity that makes the sheaf interesting in the first place.

2. **Filter onto** $\ker D_{\mathcal{F}}$**.** Apply a polynomial approximation to a projector via quantum singular value transformation, isolating the harmonic subspace.
3. **Extract a scalar.** Estimate ω, or an overlap with a prepared state, by amplitude estimation.

Remark 18.2 (Where this programme is honest and where it is not). Step 1 is the only step with genuine novelty for cellular sheaves with matrix-valued restrictions; steps 2 and 3 are standard. The honest statement of the advantage is therefore about the *encoding* and its resource scaling, not about an end-to-end speedup for TATIANA. In the deployed regime the restriction maps are the identity (Proposition 3.6), $L = L_K \otimes I_d$, and the whole object reduces to a graph Laplacian on seven vertices — for which classical computation is instantaneous and any quantum treatment is theatre. The quantum question only becomes non-trivial once Open Problem 17.1 is solved and the maps are genuinely matrix-valued and heterogeneous.

18.3 The obstruction: input and output

Claims and non-claims. Two costs dominate and both are classical.

Input. The stalk data is a set of 384-dimensional embeddings produced by a classical encoder. Preparing an amplitude encoding of these costs, in the absence of a QRAM, time linear in the data — which is the same order as computing ω classically. Any claimed speedup that assumes free state preparation is assuming away the problem.

Output. ω is one real number, and amplitude estimation returns it to precision ε in $O(1/\varepsilon)$ queries. For a diagnostic thermometer read once per tick, ε need not be small, but neither is the classical cost large. The regime where the quantum route can win is not "compute ρ faster"; it is "compute something about the spectrum of L that is classically infeasible", and no such quantity is currently required by the architecture.

18.4 What would make this real

Open Problem 18.3 (A quantity worth the encoding). Identify a functional of L that (i) the architecture genuinely needs, (ii) is classically infeasible at the scale TATIANA reaches, and (iii) is extractable from the block-encoded $D_{\mathcal{F}}$ by a constant number of scalar measurements. Without (i) the programme is solving an unposed problem; without (ii) it is slower than the classical route; without (iii) the output cost swamps the encoding.

Open Problem 18.4 (Subnormalisation in terms of sheaf data). Give an explicit bound on the block-encoding subnormalisation γ for a cellular sheaf Dirac operator in terms of the restriction-map norms and the degrees of the complex, and determine how it degrades as the maps become heterogeneous. This is the resource-scaling statement on which any advantage claim rests.

Remark 18.5 (Relation to the eddy-current programme). The block-encoding of a cellular sheaf Dirac operator with matrix-valued restriction maps is the technical core of a separate research programme on magnetoquasistatic problems, where the sheaf arises from a discretised PDE rather than from a cognitive architecture and the subnormalisation scales in material contrast. The mathematics is the same; the physical problem supplies what Open Problem 18.3 asks for and TATIANA, at present, does not. That asymmetry is the honest reason this part is short.

Appendices

Appendix A

Notation

Symbol	Meaning
$\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$	cognitive state space
$\mathcal{C}$	stratified base complex
K	coarse complex (organs as vertices)
$\mathcal{C}_v$	fine complex inside organ v
$\sigma \trianglelefteq \tau$	σ is a face of τ
$\mathcal{F}$	cellular sheaf on $\mathcal{C}$
$\mathcal{F}(\sigma)$	stalk over σ
$\mathcal{F}_{\sigma \trianglelefteq \tau}$	restriction map $\mathcal{F}(\sigma) \to \mathcal{F}(\tau)$
$C^k(\mathcal{C}; \mathcal{F})$	k-cochains $\bigoplus_{\sigma \in \mathcal{C}_k} \mathcal{F}(\sigma)$
δ	coboundary $C^0(\mathcal{C}; \mathcal{F}) \to C^1(\mathcal{C}; \mathcal{F})$
$L = \delta^\top \delta$	degree-zero sheaf Laplacian
$\omega(x) = x^\top L x$	discord (Dirichlet energy)
ρ	coherence score in $[0, 1]$
α	dissent fraction
$\tilde{\rho}$	mode index
B	Anderson–Morley normaliser
π_e	edge precision
$\perp$	undefined / measurement not made
b_0, b_1	Betti numbers of the 1-skeleton
$H^k(\mathcal{C}; \mathcal{F})$	k-th sheaf cohomology
$\mathfrak{D}$	operator algebra of growth operators
$D_{\mathcal{F}}$	sheaf Dirac operator $d_{\mathcal{F}} + d_{\mathcal{F}}^\dagger$
d	stalk dimension (384 in deployment)

Appendix B

Discrepancy Register

Places where the sources disagree with each other, or where the implementation determines a mathematical fact that the prose states differently. Each is a fact about TATIANA, not a typo.

D1 — Concepts "inside the stalk". `TATIANA_LOGBOOK.md` §0.1 describes the fine level as "concepts inside each organ's stalk". The stalk is a vector space; the fine complex is a simplicial complex. Conflating them makes the coarse-graining π_v look like an identity when it is a lossy precision-weighted fusion. Resolved in favour of the separation (Remark 2.7).

D2 — Restriction maps point the wrong way in the poster. `POSTER/sections/05_sheaf.tex` defines $\rho_{\tau\to\sigma} : \mathcal{F}(\tau) \to \mathcal{F}(\sigma)$ and draws the arrows downward, then asserts $s \in H^0(\mathcal{C};\mathcal{F})$. These are incompatible (Proposition 2.1); the downward convention defines a cosheaf. `TATIANA_Mathematical_Descript` has it correct. **Action: fix the poster before showing it.**

D3 — Two different systems named MOS. The `README` on `main` describes a retrieval pipeline (EvidenceEngine, KnowledgeCurator, Canonicalizer). The poster and the mathematical description define a sheaf-theoretic state space. These are different objects sharing a name, and the README is what a reader sees first.

D4 — `main` is stale. At the time of writing, `main` is four commits and roughly six weeks behind `fix-11-13-and-curvature-decisions`, which carries +33,450 lines including all of Part VI's material and the PRECILLA package. Any reader who clones the default branch gets a snapshot that this book does not describe.

D5 — `hodge.py` is scalar, not sheaf-valued. The Hodge decomposition code operates on a scalar complex — the trivial sheaf $\mathcal{F}(\sigma) = \mathbb{R}$ — with $(\delta^0 f)_{uv} = f(v) - f(u)$. It is therefore a computation about the *shape*, not about the sheaf. This is correct for what it measures and should not be cited as evidence about sheaf-valued cohomology.

D6 — The Hebbian-weight gap appears to be closed. An earlier scrutiny session recorded that plasticity weights were computed but never read by the coherence measurement, which would make ρ a different functional from the one documented. The current `coherence.py` computes a precision-weighted degree $d_\pi(v) = \sum_{e \ni v} \pi_e$ and a weighted Anderson–Morley bound, so the gap looks repaired. **This needs confirming against the precision source** before the finding is retired.

Appendix C

Refuted-Claims Ledger

Claims that were made, tested, and failed. They are kept because the refutations are the expensive part.

R1 — Encoder anisotropy explains the dead dynamic range of ρ. **[REFUTED]** (2026-07-28). The deployed encoder returns unit-normalised vectors, for which $\alpha = (n-1)(1-\bar{c})/n$ exactly, $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ with $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced jointly by α, by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$.

R2 — The cohomological hierarchy is informative as built. **[REFUTED]**. The 1-cochain the engine emits is δx for a global state x; a coboundary is exact, so its class in $H^1(\mathcal{C}; \mathcal{F})$ vanishes identically, for every state and every sheaf. The growth mechanism has no address in $H^1(\mathcal{C}; \mathcal{F})$. See Open Problem 17.2.

R3 — Proposition 12.4 of the earlier description. **[CONTESTED BY AUDIT]** The audit records it as false in the deployed case. The statement and the counterexample must be reproduced here before this ledger entry is complete.

R4 — Theorem 7.5 as stated. **[CONTESTED BY AUDIT]** The audit records that it is stated more generally than it is proved. The gap between hypothesis and proof must be located precisely.

R5 — K_0 classes as memory invariants. **[CONTESTED BY AUDIT]** The audit records that the K_0 classes are trivial for every object the system produces, which would mean the invariant separates nothing. See Open Problem 17.7.

Bibliography